EVQ-Cosh: Variational Frequency Allocation for Rotary Position Embedding

Anonymous Author(s)
Affiliation
Address
email

Abstract

Rotary position embedding (RoPE) is usually designed along two axes: the base frequency and inference-time position rescaling. We study a third axis, the training-time allocation of the $d_{\text{head}}/2$ frequency channels across the spectrum. We formulate this allocation as a variational inverse problem over log-frequency densities and derive EVQ-COSH, a closed-form one-parameter family in which geometric RoPE is the $\tau{=}0$ boundary under a broadband functional surrogate of the phase-collision kernel. EVQ-cosh changes only the inverse-frequency initialization and adds zero learned parameters. A semi-analytic operating-point rule, $\tau^{*}{=}d_{\text{head}}/\sqrt{L}$, selects a non-geometric allocation; its d_{head} and $L^{-1/2}$ structure is derived, and the remaining $\mathcal{O}(1)$ normalization is validated as lying inside a flat empirical PPL basin. Our primary evidence is tiered: in PE-dominant DAPE-style $128{\rightarrow}8$K extrapolation, EVQ improves over geometric RoPE and a learned positional operator; under matched YaRN scale, EVQ increases YaRN leverage on a 454M passkey-mix model, reaching 100% passkey retrieval at 8K while Geo+YaRN remains at 61%; and in compressed-RoPE MLA with 16 frequency channels, EVQ gives the largest gains. These results support training-time frequency allocation *shape*, not only frequency *range*, as a first-order RoPE design axis.

1 Introduction

Rotary position embedding (RoPE; Su et al., 2024) has become the dominant positional mechanism in modern transformers, from LLaMA-3 [Grattafiori et al., 2024] and Qwen2.5 [Yang et al., 2024a] to DeepSeek-V3 [DeepSeek-AI, 2024b] and video DiTs [Wei et al., 2025]. Two design axes are routinely tuned: the *base frequency* b is adjusted to match training length, and *inference-time position rescaling*—NTK-aware scaling [Rozière et al., 2023], positional interpolation [Chen et al., 2023], YaRN [Peng et al., 2024], and LongRoPE [Ding et al., 2024, Shang et al., 2025]—extends the usable range on a frozen checkpoint.

A third axis has received substantially less direct attention: how the $d_{\text{head}}/2$ frequency channels are *allocated* across the spectrum at training time. Since RoFormer, essentially every mainstream implementation has inherited the same geometric (equal-ratio) schedule, treated as a fixed constant rather than a variable to be chosen. Prior work has touched this axis only obliquely. Barbero et al. [2025] observe empirically that high- and low-frequency channels specialize differently after training; Wei et al. [2025] propose modality-specific allocations for video; Wang et al. [2024] preserve critical frequencies during interpolation; Shang et al. [2025] search per-channel *post-hoc* rescaling factors via evolution on a frozen checkpoint. What has been missing is a closed-form variational treatment of the *training-time* allocation itself.

We study this axis. Treating a discrete RoPE design as a density $\rho(\phi)$ over log-frequency $\phi \in [0, 1]$, we write a variational objective under a broadband functional surrogate of the phase-collision kernel

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and obtain EVQ-COSH, a closed-form one-parameter family in which geometric RoPE appears as the $\tau{=}0$ degenerate boundary. The resulting warp changes only the inverse-frequency initialization, adds no learned parameters, and is invertible in closed form. Complementing the exact variational layer, a semi-analytic rule $\tau^{*}{=}d_{\text{head}}/\sqrt{L}$ selects a robust operating point inside a wide non-geometric basin; we do not claim it is the unique optimum—its role is to move the allocation away from the degenerate $\tau{=}0$ edge rather than to pinpoint a perfect τ.

Empirical scope. We report three *primary* 3-seed settings: (P1) EVQ as a training-time substrate for YaRN in a 454M from-scratch transformer; (P2) PE-dominant extreme extrapolation (128→8K) against DAPE [Zheng et al., 2024a]; (P3) MLA [DeepSeek-AI, 2024a,b] with 16 RoPE channels at 432M, where channels are scarce and EVQ's effect is largest. We additionally report *robustness* experiments on the same MHA setting, and *supporting / exploratory* single- or two-seed evidence (video DiT, LoRA into LLaMA-3-8B, progressive training) whose role is scope-extending rather than claim-bearing. The explicit tier assignment is given in Table 3.

Contributions.

- **Axis formalization.** We formulate RoPE channel-frequency allocation as a variational problem over densities on log-frequency, and obtain a broadband collision functional whose stationary ODE admits a closed-form family.
- **Closed-form EVQ-cosh family.** We derive an invertible one-parameter allocation $\phi_k(\tau)$ in which geometric RoPE appears as the $\tau{=}0$ boundary under the proposed surrogate; the method modifies only the inverse-frequency initialization and adds no learned parameters.
- **Robust operating-point rule.** We show that a single semi-analytic rule $\tau^{*}{=}d_{\text{head}}/\sqrt{L}$—with derived d_{head} and $L^{-1/2}$ structure and an empirically calibrated $\mathcal{O}(1)$ scale—selects an allocation inside a wide non-geometric basin, within $< 1\%$ PPL of the best observed sweep points in the validated settings.
- **Empirical validation with explicit tiering.** We provide 3-seed evidence in MHA, PE-dominant extrapolation, and MLA, together with supporting single- or two-seed evidence on video DiT and LoRA at 7B; the latter are reported as preliminary.

2 Related Work

Positional encodings can be separated along three axes: the *operator* that injects positional information, the *inference-time* mechanism that extends its usable range, and the *frequency-usage* analysis of what channels actually do after training. We position EVQ-cosh in each and summarize the landscape in Table 1.

Axis A: Positional-encoding operators. Transformers rely on positional structure to recover order [Vaswani et al., 2017]; the design space spans absolute encodings, relative biases [Shaw et al., 2018, Raffel et al., 2020], additive linear biases (ALiBi; Press et al., 2022), kernelized biases (KERPLE; Chi et al., 2022), rotary position embedding (RoPE; Su et al., 2024), magnitude-decay rotary (XPOS; Sun et al., 2023), and learned positional functions (FIRE, Li et al., 2024; DAPE, Zheng et al., 2024a). EVQ-cosh is not a new operator: it preserves the RoPE rotation and modifies only how the fixed number of frequency channels is distributed along the spectrum. In the PE-dominant regime (§4.3) this zero-parameter allocation change outperforms a learned positional operator, isolating the frequency-structure effect from operator capacity.

Axis B: Inference-time and post-training context extension. NTK-aware scaling [Rozière et al., 2023], positional interpolation [Chen et al., 2023], and YaRN [Peng et al., 2024] rescale RoPE positions on a frozen checkpoint; SelfExtend [Jin et al., 2024] uses grouped attention without fine-tuning; LongRoPE [Ding et al., 2024] and PoSE [Zhu et al., 2024] use progressive or skip-wise strategies; LongRoPE2 [Shang et al., 2025] searches per-channel scaling factors by evolution on a frozen checkpoint; CLEX [Chen et al., 2024] trains a continuous extrapolation module. All of these preserve the pretrained checkpoint and modify behavior at or after inference. EVQ-cosh is orthogonal: it changes the *training-time* frequency substrate itself, and in our experiments (§4.2) composes with

Table 1: Compact comparison of RoPE-family context methods. EVQ-cosh is the only row whose main axis is closed-form variational training-time frequency allocation.

Method	Train alloc.	Infer scale	Post-hoc/ search	Learned	Closed form	Main axis
Geometric RoPE [Su et al., 2024]	fixed	—	—	—	✓	default operator
NTK / PI [Rozière et al., 2023, Chen et al., 2023]	—	✓	post-hoc	—	✓	global range
YaRN [Peng et al., 2024]	—	✓	post-hoc	—	✓	piecewise range
LongRoPE [Ding et al., 2024]	—	✓	search	—	—	range schedule
LongRoPE2 [Shang et al., 2025]	—	✓	search	—	—	per-channel range
DAPE [Zheng et al., 2024a]	—	—	training	✓	—	adaptive operator
FIRE [Li et al., 2024]	—	—	training	✓	—	learned function
VideoRoPE [Wei et al., 2025]	heuristic	✓	design	—	✓	modality-specific
Resonance RoPE [Wang et al., 2024]	—	✓	design	—	✓	critical frequencies
EVQ-cosh (ours)	✓	—	—	—	✓	**train-time allocation**

YaRN under matched YaRN scale. *Relation to LongRoPE2.* LongRoPE2 is the closest post-hoc per-channel rescaling method to EVQ, but the two target different stages: LongRoPE2 searches scaling factors for a pretrained checkpoint, whereas EVQ specifies the training-time frequency allocation before optimization begins. The methods are therefore conceptually composable; we leave empirical EVQ+LongRoPE2 composition to future work.

Axis C: Frequency-usage and allocation analysis. Barbero et al. [2025] show empirically that high-frequency channels specialize in positional patterns while low-frequency channels carry semantic content, an asymmetry that geometric allocation does not model. Wang et al. [2024] (Resonance RoPE) identify critical frequencies that should not be perturbed during interpolation, implicitly acknowledging channel inequality but framed around interpolation rather than allocation. Wei et al. [2025] propose sequence-length-dependent allocation tailored for video generation; Zhao et al. [2025] correct frequencies at inference time for video diffusion; Qiu et al. [2024] addresses temporal coherence via noise rescheduling (adjacent, not a positional-encoding method). VideoRoPE and Resonance RoPE support the premise that RoPE channels are not interchangeable, but they do not provide a closed-form variational training-time allocation for standard RoPE. We are not aware of a closed-form variational framework for training-time RoPE frequency allocation.

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024, Hsieh et al., 2024]. Prior work shows that long contexts can fail even when relevant information is nominally present [Liu et al., 2024], and that positional encoding modifications can mitigate such failures [Zhang et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

Positioning. EVQ-cosh occupies a previously unfilled cell (Table 1): a training-time, closed-form, zero-parameter frequency-allocation substrate. It is compositional with every method in Axis B—it supplies the frequency structure the others then rescale or extend—and is complementary to the adaptive-operator methods in Axis A.

3 Theory and Method

Before any derivation we state the epistemic status of each component (Table 2), so that the variational core is not conflated with the practical recipe. In our view this separation is more important than any particular identity in the proofs below.

The surrogate is functional (not pointwise: Appendix A.4 reports 24–92% collision-score reduction under the exact kernel across 12 configurations). The pure-tether branch of the stationary ODE (Theorem 1) is a design choice that keeps the allocation invertible in closed form; it is not a claim that the Fisher-driven forcing term never matters. The rule $\tau^{*}{=}d_{\text{head}}/\sqrt{L}$ (Proposition 1) is an operating-point selector whose d_{head} factor and $L^{-1/2}$ exponent are derived; its remaining $\mathcal{O}(1)$ scale is fixed by normalization convention, and its role is to move the allocation away from the $\tau{=}0$ boundary into a wide non-geometric basin.

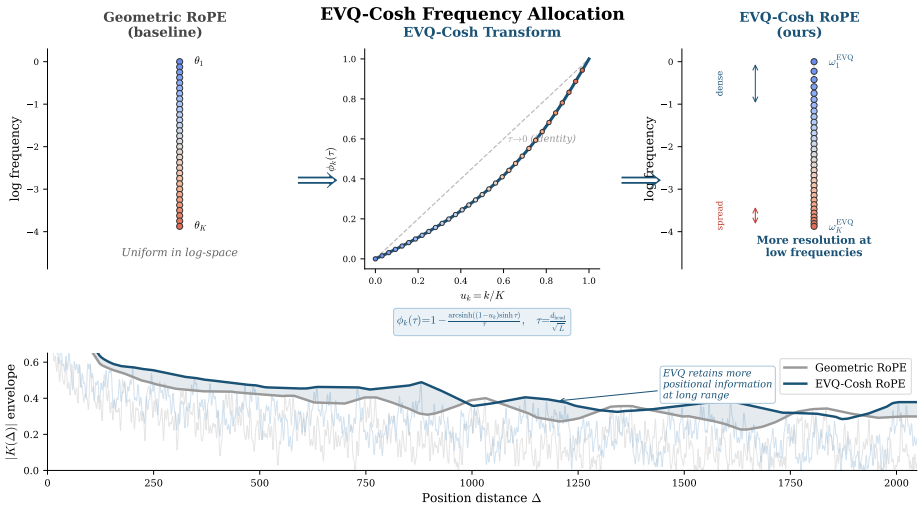

Figure 1: **EVQ-Cosh frequency allocation.** Left: geometric RoPE distributes frequencies uniformly in log-space. Center: the EVQ-cosh warp $\phi_k(\tau)$ bends the allocation below the identity (dashed), concentrating resolution at low frequencies; $\tau \to 0$ recovers geometric RoPE. Right: the resulting allocation spreads low-frequency channels, retaining more positional information at long range.

Table 2: Epistemic map of the EVQ-cosh derivation. The variational core (rows 1–2) is exact conditional on the broadband surrogate; the practical recipe (rows 3–4) is a deliberate engineering choice whose residual uncertainty is bounded empirically (§3.7).

Component	Status	Role
Broadband surrogate K_{app}	Functional approximation of the *collision objective* (not pointwise approximation of the kernel)	Yields a tractable variational problem; validated in Appendix A.4
ODE and closed-form family	Exact conditional on K_{app}	Theoretical core; geometric RoPE is the $\tau{=}0$ boundary
Pure-tether branch	Practical branch choice of the stationary ODE	Gives a closed-form invertible quantile warp; forcing term suppressed at typical bases
τ^* scaling rule	Semi-analytic operating-point selector (not a global minimizer)	Moves allocation away from the degenerate $\tau{=}0$ edge into a wide non-geometric basin

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is equivalent to choosing a density $\rho(\phi)$ over this interval and discretizing it by inverse CDF (Figure 1). Geometric RoPE corresponds to uniform ρ; EVQ derives ρ from a variational objective. This reparameterization lifts the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

The collision objective admits a broadband functional surrogate,

$$K_{\text{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance. The δ term captures local ridge structure; $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. The exact kernel is oscillatory and cannot be pointwise-matched by a smooth two-parameter model; K_{app}

captures the *functional* structure of the collision objective. Appendix A.4 validates this: the EVQ solution derived from K_{app} reduces the collision score under the exact kernel by 24–92% across 12 configurations, and a collision-only numerical oracle search (Appendix A.12) reaches comparable order-of-magnitude reductions, suggesting that non-cosh families may further reduce exact-kernel collision at the cost of losing closed-form invertibility. A self-consistency theorem (Theorem 3 in Appendix) further confirms the surrogate's structural rigidity.

3.3 Variational objective and stationary ODE

The broadband functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int \rho(\phi)^2\, d\phi + \frac{\beta}{2}\iint \rho(\phi)\rho(\psi)\min(\phi,\psi)\, d\phi\, d\psi - \mu\int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1 (Stationary ODE)**.** *Conditional on K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2\rho(\phi) = \gamma\, b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

The ODE and its closed-form family are exact conditional on K_{app}, with integration constants fixed by *natural* boundary conditions from the $\min$-kernel Green structure (Appendix A)—not by anchoring endpoint frequencies. The decomposition separates a global tether (redistributing density) from a local Fisher-utility forcing term, ensuring EVQ is the stationary compromise rather than an unconstrained long-range distortion.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,, \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2 (Geometric limit)**.** *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Geometric RoPE is therefore the $\tau{=}0$ degenerate limit of the EVQ family, not a generic optimum.*

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the complete implementation is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

The construction is deliberately minimal: set τ, compute quantiles, evaluate the warp, replace inverse frequencies. No auxiliary loss, no extra parameters, no optimizer dependence. The endpoint grid $u_k{=}k/K$ has the anchoring property $u_0{=}0{\Rightarrow}\phi_0{=}0$ (preserves $\omega_0{=}1$); our implementation uses midpoint quantization $u_k{=}(2k{+}1)/(2K)$, which trades exact anchoring for avoiding endpoint cells. We do not rely on $\omega_0{=}1$ in any trained model, and the endpoint/midpoint choice affects PPL by $<1\%$ in the $<1\%$ PPL basin across all $K{\geq}16$ settings we tested.

EVQ-cosh changes only the inverse-frequency initialization—distinct from learnable PE approaches that rely on in-distribution training dynamics to discover the allocation through extra parameters. The full recipe is Equation (6) above plus $\tau{=}d_{\text{head}}/\sqrt{L_{\text{train}}}$: zero learned parameters, architecture unchanged.

3.6 Waterbed inequality

Under the broadband surrogate, EVQ obeys a tight waterbed inequality (Proposition 3, Appendix A.6): the error functional $\mathcal{W}(\rho) = \alpha\chi^2(U\|P_\rho) + \beta\int M_\rho^2$ satisfies $\mathcal{W}(\rho) \geq \alpha(e^{D_B(\rho)} - 1)$ with equality iff $\rho \equiv 1$, and EVQ's in-range cost scales as $\mathcal{W}(\rho_{\tau_*}) = O(d_{\text{head}}^4/(L_{\text{eff}}^J)^2)$. Long-range improvement is thus paid for by a bounded, quartic-in-τ^* in-range cost—not eliminated.

3.7 The τ^* scaling law: epistemic stratification

Shape and scale come from separate layers. The cosh shape is the pre-softmax variational optimum under K_{app} (§3.3–§3.5); the scale τ^* is then selected *post*-softmax by the following proposition, whose $1/L$ factor is the softmax Jacobian eigenvalue at the diffuse long-range baseline $p_0{=}1/L$—an algebraic property of softmax, not a CLT/i.i.d. claim about token sequences.

Proposition 1 (Softmax Transport Temperature Selection)**.** *Define $\mathcal{F}(\tau) = S_{\chi^2}(\tau) - \lambda\, U(\tau, L)$, where $S_{\chi^2}(\tau)$ is the Pearson χ^2 stiffness and $U(\tau, L) = (d_{\text{head}}/L) \int_0^1 q(Lb^{-\phi})\rho_\tau(\phi)d\phi$ is the post-softmax transport utility, $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$ (Appendix A.10). The $1/L$ factor arises from the softmax Jacobian at $p_0 = 1/L$; d_{head} is the per-head normalization shared by stiffness and utility. In the small-τ limit, the minimizer satisfies*

$$\tau_*^2 \;=\; 45\,\lambda\, Q_1(L,b)\cdot \frac{d_{\text{head}}^2}{L}, \qquad Q_1(L,b) \;=\; \int_0^1 \eta(\phi)\, q(Lb^{-\phi})\, d\phi,$$

with $\eta(\phi) = (1-\phi)^2/2 - 1/6$, hence $\tau_* \propto d_{\text{head}}/\sqrt{L}$ with L-exponent exactly $-1/2$.

Non-diffuse baseline and L_{eff}. The $1/L$ factor used $p_0{=}1/L$. Under a trained attention distribution the corresponding scalar is $1/L^J_{\text{eff}}$, the reciprocal softmax-Jacobian Rayleigh quotient on the RoPE-induced logit transport direction g_i; g_i has a closed form (39) and L^J_{eff} is defined as a ratio-of-means (41) (Appendix A.13). This gives $\tau^* \propto d_{\text{head}}/\sqrt{L^J_{\text{eff}}}$, and writing $L^J_{\text{eff}}{=}\Theta(L^a_{\text{train}})$ unifies the rule as $\tau^* \propto L_{\text{train}}^{-a/2}$, with $a{\approx}1$ on our $L{\leq}8$K grid. A cheap entropy proxy $L^H_{\text{eff}}{=}(\mathbb{E}[e^{-H}])^{-1}$ agrees with L^J_{eff} under flat-window attention but diverges at near-one-hot heads (where softmax saturates), so we use L^H_{eff} only as a diagnostic. Saturation ($a{\to}0$) is empirically testable and would revise τ^* at $L_{\text{train}}{\geq}16$K.

Three epistemic tiers. The d_{head} and $L^{-1/2}$ structure is **derived** (exact in the small-τ limit; at practical τ numerical optimization yields $\gamma \approx 0.465$, within 1% PPL of $L^{-0.500}$). The χ^2 stiffness is **first-principles under a channel-load axiom**: with channels sampled according to $\rho\, d\phi$ and local cell width $\ell(\phi){=}1/\rho(\phi)$, $S_{\chi^2}(\rho) = \text{Var}_\rho(1/\rho)$, i.e. the variance of per-channel quantization load (Appendix A.15). Among load-moment penalties $S_p(\rho){=}\mathbb{E}_\rho[(1/\rho - 1)^p]$ only $p{=}2$ reproduces $L^{-1/2}$; smooth f-divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho{=}1$ and agree with χ^2 at leading order, but their *finite-τ* best points over our L grid give L-exponents $\gamma \in \{0.58\text{–}0.81\}$—outside the $\pm1\%$ PPL basin around $\gamma{=}0.500$ at the tested scales (Table 10). The Lagrange multiplier λ is an $\mathcal{O}(1)$ **curvature-ratio** between the surrogate and exact utilities (Appendix A.16); the collision-score oracle yields $c_{\text{coll}} = 1.171 \pm 0.001$ (CV 0.11%, 9 configurations; Table 8), which we read as a 17% surrogate curvature under-estimate rather than as a free fit parameter; using $\lambda{=}1$ remains inside the $<1\%$ PPL basin across all 27 configurations.

Basin, not point estimate; additive corrections. With $\lambda{=}1$, the formula falls within $< 1\%$ PPL of the best observed sweep points across 27 configurations (MHA 50M–750M, MLA, video DiT, post-hoc LoRA at 7B; Appendix A.9): a wide near-optimal basin surrounds the variational solution while the uniform $\tau{=}0$ baseline sits outside it, so the derivation identifies the *direction*, not a perfect τ. EVQ corrects *shape* while YaRN corrects *range* (Figure 2); the corrections are additive, predicting the strong EVQ+YaRN composition observed in §4, and attention-distance measurements verify the waterbed crossover at $\approx$508 tokens (Figure 6, Appendix).

4 Experiments

4.1 Setup and evidence tiering

Experiments test three questions derived from the variational characterization: (Q1) does EVQ improve over Geo at fixed training length? (Q2) does EVQ compose with inference-time scaling, i.e., does EVQ+YaRN outperform Geo+YaRN at extrapolated lengths under matched YaRN scale? (Q3) does the τ^* rule land in the best observed sweep basin across architectures and scales? Core experiments train from scratch with fixed architecture, modifying only the RoPE inverse-frequency initialization. We use $b{=}500$K, the predicted $\tau^*(L)$ for each training length, and standard GPT-style

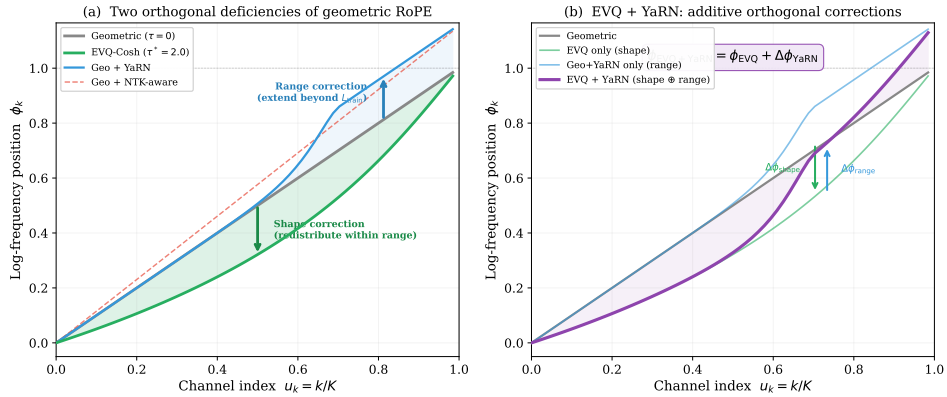

Figure 2: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** (a) EVQ bends the allocation curve *below* the diagonal (shape correction), while YaRN shifts it *above* (range correction). (b) EVQ+YaRN combines both corrections additively. Setup: base $= 10{,}000$, d_{head}=128, L_{train}=4096, $4\times$ extrapolation.

decoder stacks [Radford et al., 2019] with d_{head}=64 across 50M–750M. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; evaluations target PE-diagnostic metrics (extrapolation PPL, passkey retrieval, NIAH, downstream NLL) that isolate positional encoding quality from model capacity. Full hyperparameters are in Appendix D.

Evidence is organized in three tiers (Table 3). *Primary* (3-seed, directly tied to central claims): EVQ$\times$YaRN composition (§4.2), PE-dominant extrapolation (§4.3), and MLA (§4.4; full table in Appendix C.1). *Robustness* (3-seed, single architecture): FineWeb-Edu consistency, base sweep consistency, and capability preservation (§4.5). *Supporting / exploratory* (1–2 seed): the one-seed rows in the multi-scale raw-PPL table, video DiT, LoRA into LLaMA-3-8B, progressive training, and the MLA 1B-token run (§4.6). No claim made in the abstract or introduction rests on tier-3 evidence.

Table 3: Evidence tier by setting. Primary claims rest only on 3-seed primary tier; robustness rows are 3-seed but not primary anchors.

Tier	Setting	Seeds	Location
Primary I	EVQ+YaRN, 454M MHA, 4K→8/12/16K	3	§4.2
Primary II	PE-dominant 128→8K (DAPE comparison)	3	§4.3
Primary III	MLA 432M, 16 RoPE channels, 500M tokens	3	§4.4
Robustness	FineWeb-Edu consistency, base sweep, capability preservation	3	§4.5
Supporting	Multi-scale raw-PPL one-seed rows	1	§4.5, App. D
Supporting	Video DiT 129M / 382M	1–2	§4.6, App. D
Supporting	LoRA into LLaMA-3-8B-Instruct	1	§4.6, App. D
Supporting	Progressive training	1	§4.6, App. D
Supporting	MLA 1B-token (robustness to training saturation)	1	§4.4, App. D

4.2 Primary I: EVQ increases YaRN leverage under matched-scale scaling

Figure 3 and Table 4 present the paper's main systems result. The decisive comparison is EVQ+YaRN against Geo+YaRN under the same YaRN scale. This fixed-scale protocol is a fairness control that isolates the training-time frequency substrate; it is not a claim that YaRN itself is weak or untunable.

In the 454M 10% passkey-mix setting (3 seeds per configuration), EVQ+YaRN reaches 100% passkey retrieval at 8K while Geo+YaRN remains at 61%; the advantage persists at longer contexts, with EVQ+YaRN at 79%/68% versus 59%/51% for Geo+YaRN at 12/16K (Table 4). The same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. EVQ alone improves over Geo without saturating retrieval, so the jump to 100% under YaRN is not a ceiling effect. The composition reflects the orthogonal deficiency structure: EVQ corrects *shape*, YaRN corrects *range*, and their corrections are additive (Figure 2).

Table 5 quantifies the interaction under matched YaRN scale: YaRN provides only $-4.5\%/-3.1\%$ on Geo (4/8K) but $-32.5\%/-40.7\%$ on EVQ (τ=4). NTK-aware scaling becomes destructive

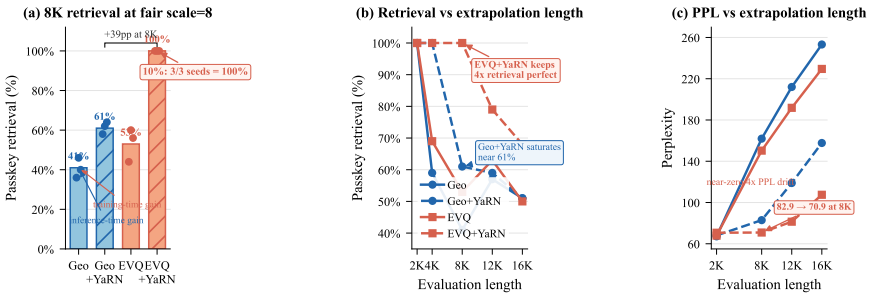

Figure 3: EVQ and YaRN correct orthogonal deficiencies. EVQ expands low-frequency spacing at training time; YaRN rescales positions at inference time. Their combination under matched YaRN scale substantially outperforms Geo+YaRN on both retrieval and PPL.

Table 4: Training-time versus inference-time positional optimization (454M, L_{train}=512, 10% passkey mix). Evaluation at 8/12/16K corresponds to $16\times/24\times/32\times$ extrapolation relative to training length. YaRN is applied with the *same fixed* scale s=4 to Geo and EVQ (not the standard $s=L/L_{\text{train}}$); this is a matched-scale control protocol for isolating the training-time frequency substrate, not an assessment of YaRN scale tuning. Under this identical-scale comparison, EVQ+YaRN reaches 100% passkey retrieval at 8K while Geo+YaRN plateaus at 61%, and the advantage persists at 12/16K (79%/68% vs. 59%/51%). PPL values use full-sequence scoring on held-out FineWeb-Edu; see Table 24 for per-document scoring on the same model.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

at large τ, consistent with the view that complementarity requires preserving the EVQ frequency structure.

We read this as EVQ substantially increasing YaRN's leverage in matched-scale settings; a per-checkpoint tuned-scale YaRN comparison is deferred, and we do not claim dominance over every tuned YaRN baseline.

4.3 Primary II: PE-dominant extreme extrapolation

Table 6 provides the second principal anchor, isolating frequency allocation from model capacity. In DAPE-style 128→8K extrapolation (reproducing the extreme-extrapolation setting of Zheng et al. [2024a]), EVQ achieves 333.7 PPL (-35% vs. the Geo baseline 513.7), surpassing DAPE (455.3) with zero extra parameters. At L=256, the predicted τ^*=4.0 is preferred across both 125M and 454M—scale changes extrapolation difficulty but not the preferred allocation (visualized in Appendix Figure 10). In this setting, the model has minimal capacity to absorb allocation errors, so the frequency structure itself is the binding variable: a zero-parameter allocation rule outperforms a 32-parameter learned positional operator.

4.4 Primary III: MLA with scarce RoPE channels

Multi-head Latent Attention (MLA) [DeepSeek-AI, 2024a,b] carries positional information in only d_{rope}=32 dimensions—16 frequency channels after pairing—versus 64 in standard MHA with d_{head}=128. On a 432M MLA transformer trained for 500M tokens at L_{train}=8192 (3 seeds), EVQ alone reduces $2\times$ extrapolation PPL from 138.8 to 95.6 (-31.1%) with $+1.1\%$ in-distribution cost, surpassing Geo+YaRN (117.9; Appendix C.1). The ranking EVQ+YaRN > EVQ > Geo+YaRN > Geo holds at every tested ratio, and the 31.1% gain exceeds typical MHA gains (13–15%)—consistent with the prediction that allocation quality becomes decisive as channels compress.

Table 5: Matched-scale YaRN leverage at $L = 256$; NTK-aware scaling is not complementary at large τ.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

Table 6: PE-dominant summary (125M, 3-seed mean): EVQ beats learned PE at $128 \to 8$K, and $\tau = 4.0$ combines strongly with YaRN at $L = 256$.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

Robustness to training saturation (preliminary). Under fully-trained MLA (1B tokens, single seed; Appendix D), EVQ's *standalone* advantage narrows and reverses on some in-range metrics, but EVQ+YaRN after fine-tuning still improves over Geo+YaRN+FT by -2.5%. We report this as preliminary robustness evidence that training-time allocation remains the binding constraint for composition with inference-time methods, rather than as a conclusive training-saturation story in itself.

4.5 Robustness

Multi-scale raw extrapolation (Table 21, Appendix) separates the 3-seed FineWeb-Edu robustness row from the one-seed supporting rows: in-range PPL stays within $\pm 1.7\%$ while long-range PPL improves by 10–46%. Base sweeps at b=10K/500K and capability checks (Table 24) further confirm EVQ is non-destructive on tasks without long-range retrieval; none of these rows is a primary claim on its own.

4.6 Supporting and exploratory evidence

Supporting rows extend scope but carry no abstract or introduction claim: video DiT (129M / 382M, 1–2 seeds) treats the $0.53\times$ τ correction as rationalization; LoRA into LLaMA-3-8B-Instruct at $r{=}d_{\text{head}}/2$ recovers 8–19× extrapolation-PPL but is not a capability benchmark; and progressive training is single-seed. Full disclosures are in Appendix D.

5 Limitations and Scope

The variational derivation is exact conditional on the broadband surrogate K_{app}; the practical τ^* rule is semi-analytic: the d_{head} factor, $L^{-1/2}$ exponent, and χ^2 stiffness are derived, while the remaining $\mathcal{O}(1)$ scale is fixed by normalization convention and falls inside the observed flat PPL basin. The causal evidence is PE-diagnostic (extrapolation PPL, passkey, NIAH, gold-answer NLL); MMLU, HumanEval, RULER, and QA are useful non-regression checks, not the causal scope of the paper. From-scratch $\geq$1B text validation and production MLA at b=10K remain future work; LoRA, progressive training, video DiT 382M, and MLA 1B-token composition are supporting boundary checks. The video DiT $0.53\times$ correction is post-hoc rationalization, and a principled modality-aware τ^* is future work.

Conclusion. EVQ-cosh identifies geometric RoPE as the τ=0 boundary of a closed-form training-time allocation family; the matched-scale and compressed-RoPE results support allocation *shape*, not just *range*, as a binding RoPE design axis.

References

Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025.

Guanzheng Chen, Xin Li, Zaiqiao Meng, Shangsong Liang, and Lidong Bing. CLEX: Continuous length extrapolation for large language models. In *International Conference on Learning Representations*, 2024.

Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023.

Ta-Chung Chi, Ting-Han Fan, Peter J. Ramadge, and Alexander I. Rudnicky. KERPLE: Kernelized relative positional embedding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a.

DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.

Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, 2024.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.

Hongye Jin, Xiaotian Han, Jingfeng Yang, Zhimeng Jiang, Zirui Liu, Chia-Yuan Chang, Huiyuan Chen, and Xia Hu. LLM maybe LongLM: SelfExtend LLM context window without tuning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pages 22099–22114, 2024.

Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, et al. HunyuanVideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024.

Shanda Li, Chong You, Guru Guruganesh, Joshua Ainslie, Santiago Ontanon, Manzil Zaheer, Sumit Sanghai, Yiming Yang, Sanjiv Kumar, and Srinadh Bhojanapalli. Functional interpolation for relative positions improves long context transformers. In *International Conference on Learning Representations*, 2024.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

Xin Ma, Yaohui Wang, Gengyun Jia, Xinyuan Chen, Ziwei Liu, Yuan-Fang Li, Cunjian Chen, and Yu Qiao. Latte: Latent diffusion transformer for video generation. *arXiv preprint arXiv:2401.03048*, 2024.

Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024.

Ofir Press, Noah A. Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*, 2022.

Haonan Qiu, Menghan Xia, Yong Zhang, Yingqing He, Xintao Wang, Ying Shan, and Ziwei Liu. FreeNoise: Tuning-free longer video diffusion via noise rescheduling. In *International Conference on Learning Representations*, 2024.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Technical Report*, 2019.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, et al. Code llama: Open foundation models for code. *arXiv preprint arXiv:2308.12950*, 2023.

Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2018.

Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pages 843–852, 2015.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

Yutao Sun, Li Dong, Barun Patra, Shuming Ma, Shaohan Huang, Alon Benhaim, Vishrav Chaudhary, Xia Song, and Furu Wei. A length-extrapolatable transformer. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Suyuchen Wang, Ivan Kobyzev, Peng Lu, Mehdi Rezagholizadeh, and Bang Liu. Resonance RoPE: Improving context length generalization of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for good video rotary position embedding? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.

Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. CogVideoX: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024b.

Zhenyu Zhang, Runjin Chen, Shiwei Liu, Zhewei Yao, Olatunji Ruwase, Beidi Chen, Xiaoxia Wu, and Zhangyang Wang. Found in the middle: How language models use long contexts better via plug-and-play positional encoding. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

Chuanyang Zheng, Yihang Gao, Han Shi, Minbin Huang, Jingyao Li, Jing Xiong, Xiaozhe Ren, Michael K. Ng, Xin Jiang, Zhenguo Li, and Yu Li. DAPE: Data-adaptive positional encoding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 37, 2024a.

Zangwei Zheng, Xiangyu Peng, Tianji Yang, Chenhui Shen, Shenggui Li, Hongxin Liu, Yukun Zhou, Tianyi Li, and Yang You. Open-Sora: Democratizing efficient video production for all. *arXiv preprint arXiv:2412.20404*, 2024b.

Dawei Zhu, Nan Yang, Liang Wang, Yifan Song, Wenhao Wu, Furu Wei, and Sujian Li. PoSE: Efficient context window extension of LLMs via positional skip-wise training. In *International Conference on Learning Representations*, 2024.

Broader Impact

This work studies frequency allocation as a design axis for rotary position embedding rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing the default geometric frequency schedule with a principled variational allocation can reduce wasted compute and improve the reliability of long-context models across architectures. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{7}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{8}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{9}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{10}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{11}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{12}$$

$$g''(\phi) = -\rho(\phi). \tag{13}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{14}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{15}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{16}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{17}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{18}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{19}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether $(\cosh, \sinh)$ and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) The forcing branch is globally small in L^1/CDF transport, not pointwise. Once $\gamma(b) = (\lambda_F/\alpha)\, \bar{\eta}_F\, (2 \log b)^2$ is derived (Appendix A.14, Eq. (45)), the $(s^2 - \tau^2)^{-1}$ denominator in the particular solution $\delta\rho_F^{\text{part}} = \gamma(b)\, q_s/(s^2 - \tau^2)$ is cancelled by the s^2 in $\gamma(b)$, so the forcing amplitude is *not* pointwise $O(1/\log^2 b)$. The true suppression is boundary-layer / L^1-level: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$, giving $\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 = O(1/\log b)$ and CDF-level error of the same order (Eq. (47)). Empirically, the full solution improves over pure tether by 2–6% at unit-convention λ=1 (vs. 28–92% improvement over geometric), and the 99-run Phase 16 sweep (Figure 4) confirms that the pure-tether family matches the scaling-law prediction, subsuming geometric RoPE as its $\tau \to 0$ limit.

A.4 Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory's sole approximation. We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$ — which is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K?

Table 7 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in \{128, \ldots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in \{16, 32\}$), covering both text and video regimes.

EVQ reduces collision under the exact kernel in all 12 configurations (-24% to -92%), with effective rank improvements of $+24\%$ to $+570\%$. Two patterns emerge. First, the collision reduction monotonically increases as the geometric allocation becomes more degenerate: shorter L (more extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE (L=4096, where baseline effective rank is already 6.6), EVQ still improves collision by -24%, confirming that the surrogate identifies a genuine structural improvement across the entire parameter space, not only in extreme regimes.

Table 7: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L, larger b).

	Collision C		Eff. rank	
Configuration	GEO	EVQ (Δ_C)	GEO	EVQ
Text: K=32, b=500K, *sweep* L				
L=128	226.3	17.7 (−92%)	2.5	17.0
L=256	192.6	27.6 (−86%)	3.0	13.9
L=512	163.2	43.3 (−73%)	3.6	10.8
L=1024	133.9	59.0 (−56%)	4.4	8.8
L=2048	109.0	67.5 (−38%)	5.4	8.0
L=4096	87.0	66.3 (−24%)	6.6	8.1
Text: K=32, L=2048, *sweep* b				
b=10K	34.8	19.2 (−45%)	13.0	17.3
b=100K	79.4	47.0 (−41%)	7.1	10.4
Video: K=16, L=32, *sweep* b				
b=100	22.6	14.2 (−37%)	4.8	6.4
b=1K	42.3	28.0 (−34%)	2.8	3.9
b=10K	55.4	39.8 (−28%)	2.2	2.9
b=50K	63.3	47.2 (−26%)	2.0	2.5

A.5 Exact higher-order displacement floor

The one-channel mid-displacement threshold admits an exact Taylor series in $1/K$ that is fully closed-form (no empirical calibration).

Proposition 2 (Higher-order floor)**.** *For the EVQ-cosh mapping with $K = d_{\text{head}}/2$ channels, the N-channel mid-displacement threshold satisfies*

$$\tau_{\text{floor}}(N, K) \;=\; 4\sqrt{\frac{N}{K}}\left[1 + \frac{N}{2K} + \frac{241}{120}\left(\frac{N}{K}\right)^2 + O(K^{-3})\right]. \tag{20}$$

All coefficients are derived from the Taylor series $\Delta\varphi(1/2, \tau) = -\tau^2/16 + \tau^4/256 + 17\tau^6/30720 + O(\tau^8)$, whose coefficients $(-1/16,\ +1/256,\ +17/30720)$ are exact rationals from the expansion of $\operatorname{arcsinh}(\sinh\tau/2)/\tau$ (Appendix script `theory_B_floor_higher_order.py`; verified symbolically).

Accuracy. Against the exact floor (obtained by numerically solving $|\Delta\varphi(1/2, \tau)| = N/K$): the leading term $4\sqrt{N/K}$ has relative error 3.85% at K=16 and 0.20% at K=512; the two-term expansion reduces this to 0.088% at K=16 and below 10^{-5} for $K \geq 128$. The formula is thus a closed-form refinement of the Habitable Zone lower bound, with all constants derived from the cosh geometry alone.

A.6 Waterbed as a tight functional inequality

Setup. With $\rho > 0$, $\int_0^1 \rho = 1$, $U(d\phi) = d\phi$, $P_\rho(d\phi) = \rho\, d\phi$, $\delta\rho = \rho - 1$, and $M_\rho(s) = \int_s^1 \delta\rho\, du$, define the waterbed functional

$$\mathcal{W}(\rho) \;\coloneqq\; \alpha\, \chi^2(U \| P_\rho) + \beta \int_0^1 M_\rho(s)^2\, ds, \qquad \chi^2(U \| P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho}\, d\phi. \tag{21}$$

The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the integrated squared cumulative error, which arises from the $\min$-kernel off-diagonal covariance in K_{app}.

Proposition 3 (Waterbed inequality)**.** *Let $D_B(\rho) = -\int_0^1 \log \rho(\phi)\, d\phi$ be the Burg entropy deficit. Then*

$$\mathcal{W}(\rho) \;\geq\; \alpha\, F_*(D_B(\rho)), \qquad F_*(d) \;=\; \inf_{\rho > 0,\, \int \rho = 1,\, -\int \log \rho = d} \left[\int_0^1 \frac{1}{\rho} - 1\right], \tag{22}$$

with equality $\mathcal{W}(\rho) = 0$ if and only if $\rho \equiv 1$ (geometric RoPE). A readable corollary via Rényi monotonicity is

$$\mathcal{W}(\rho) \;\geq\; \alpha\big(e^{D_B(\rho)} - 1\big). \tag{23}$$

Proof sketch. Since $\beta \int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha\chi^2(U\|P_\rho)$. For fixed $D_B(\rho){=}d$, the sharp lower envelope of χ^2 over all densities is the Csiszár–Vajda f-divergence comparison F_*, whose extremizer is a two-level density $X = 1/\rho \in \{a, c\}$ with mass-normalization and entropy-deficit constraints. For the corollary, $1 + \chi^2(U\|P_\rho) = \int 1/\rho\, \mathbb{E}_U[\cdot] \geq e^{D_1(U\|P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality at $\rho{\equiv}1$ is immediate from both sides vanishing. □

EVQ evaluation. For $\rho_\tau(\phi) = \tau\cosh(\tau(1{-}\phi))/\sinh\tau$,

$$D_B(\rho_\tau) \;=\; \log\tfrac{\sinh\tau}{\tau} - \tfrac{1}{\tau}\int_0^\tau \log\cosh y\, dy \;=\; \tfrac{\tau^4}{90} - \tfrac{8\tau^6}{2835} + O(\tau^8), \tag{24}$$

$$S_{\chi^2}(\rho_\tau) \;=\; \tfrac{\sinh\tau}{\tau^2}\arctan(\sinh\tau) - 1 \;=\; \tfrac{\tau^4}{45} - \tfrac{2\tau^6}{315} + O(\tau^8), \tag{25}$$

$$\int_0^1 M_{\rho_\tau}(s)^2\, ds \;=\; \tfrac{2\tau^4}{945} + O(\tau^6). \tag{26}$$

Combining,

$$\mathcal{W}(\rho_\tau) \;=\; \left(\frac{\alpha}{45} + \frac{2\beta}{945}\right)\tau^4 + O(\tau^6), \qquad \mathcal{W}(\rho_{\tau_*}) \;=\; O\!\left(\frac{d_{\text{head}}^4}{(L_{\text{eff}}^J)^2}\right). \tag{27}$$

The in-range surrogate cost for EVQ is $O(\tau^4)$; since $\tau_*^2 \asymp d_{\text{head}}^2/L_{\text{eff}}^J$, the cost scales as $d_{\text{head}}^4/(L_{\text{eff}}^J)^2$. This is the explicit form of "bounded in-range cost" cited in the main text.

A.7 Surrogate self-consistency theorem

The surrogate functional $\mathcal{J}[\rho]$ admits three exact identities that connect its two integral terms $T_1(\tau) = \int_0^1 \rho^2\, d\phi$ and $T_2(\tau) = \int_0^1\!\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\, d\phi\, d\psi$, where $\rho(\phi) = \tau\cosh(\tau(1{-}\phi))/\sinh\tau$ is the normalized cosh density.

Theorem 3 (Surrogate self-consistency)**.** *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau\coth\tau. \tag{28}$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the $\min$ kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2\rho$ this gives $h'' = -\tau^2\rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$. Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2\rho - (-\tau^2\rho) = 0$, so f is linear in ϕ. The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \tag{29}$$

Integrating against ρ:

$$T_2 = \int_0^1 \rho(\phi)\, g(\phi)\, d\phi = \frac{1}{\tau^2}\big[\rho(0)\underbrace{\int_0^1 \rho}_{=1} - T_1\big] = \frac{\rho(0) - T_1}{\tau^2}. \tag{30}$$

Since $\rho(0) = \tau\cosh\tau/\sinh\tau = \tau\coth\tau$, rearranging gives (28). □

An immediate corollary follows.

Corollary 1 (Closed-form T_2)**.**

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau\sinh^2\tau}. \tag{31}$$

This follows from substituting the known closed form $T_1(\tau) = \tau(1 + \sinh 2\tau/2\tau)/(2\sinh^2\tau)$ and $\rho(0) = \tau\coth\tau$ into the theorem. Both identities are verified numerically to machine precision ($< 10^{-15}$ relative error across $\tau \in [0.01, 20]$; code: `scripts/lambda_identity.py`).

The self-consistency theorem reveals that the surrogate's two energy terms are not independent: they are locked by the cosh density's self-reproducing property under the min-kernel Green's function. This structural rigidity is why the pure-tether solution (§A) performs so well despite being a single-parameter family.

A.8 Cross-validation of λ

The scaling law $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$ is validated at two levels.

Collision-score level. We minimize the exact-kernel collision score $C(\tau)$ for each of 9 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, 512, 1024\}$) and compute $c_{\text{coll}} = \tau^*_{\text{coll}}/(d_{\text{head}}/\sqrt{L})$. The resulting $c_{\text{coll}} \approx 1.17$ is nearly constant (CV: 0.11%); leave-one-out cross-validation confirms max prediction error $< 0.4\%$ (Table 8). This near-zero spread reflects the smoothness of the collision score as a function of τ, not PPL variability.

Table 8: Collision-score cross-validation of the operating-point prefactor c_{coll} in $\tau^* = c_{\text{coll}} \cdot d_{\text{head}}/\sqrt{L}$. Each τ^*_{coll} is the exact-kernel collision-score minimizer; each row holds out one configuration and predicts τ^*_{coll} from the remaining 8. Max error 0.008 ($< 0.4\%$ relative). We reserve λ for the curvature multiplier in $\mathcal{F}{=}S_{\chi^2} - \lambda U$ (Appendix A.16); the empirical prefactor c_{coll} is related to λ by $c^2 \propto \lambda Q_1$ under the paper's normalization, so $c_{\text{coll}}{=}1.171$ reports a prefactor estimate, not λ directly. The deployed formula uses unit-convention $\lambda{=}1$, and the corresponding $c{=}1$ falls inside the flat PPL basin ($< 1\%$ gap from the best observed PPL sweep points across 27 configurations).

d_{head}	L	τ^*_{coll}	τ^*_{pred}	Error
32	256	2.340	2.342	+0.002
32	512	1.660	1.655	−0.005
32	1024	1.170	1.171	+0.001
64	256	4.680	4.684	+0.004
64	512	3.310	3.312	+0.002
64	1024	2.340	2.342	+0.002
128	256	9.370	9.366	−0.004
128	512	6.630	6.622	−0.008
128	1024	4.680	4.684	+0.004
Mean $c_{\text{coll}} = 1.171 \pm 0.001$ (CV: 0.11%); RMSE $= 0.004$				

PPL-sweep level. Phase 16 training sweeps (99 runs, 9 configurations $\times$ 3+ seeds) identify the best observed PPL sweep point for each configuration. The resulting $\lambda_{\text{PPL}} = \tau^*_{\text{PPL}}/(d_{\text{head}}/\sqrt{L})$ has mean $\approx$1.19 with broader scatter (CV $\approx$14%), because PPL depends on training noise, finite data, and the discrete sweep grid. The deployed formula uses $\lambda{=}1$ (i.e. $\tau = d_{\text{head}}/\sqrt{L}$ without a multiplicative constant), which consistently falls inside the flat PPL basin: the PPL gap between the formula and the best observed grid point is $<1\%$ across all 27 validated configurations.

Basin flatness. The collision-score landscape $C(\tau)$ for a representative configuration ($d_{\text{head}}{=}64$, $L{=}512$, $b{=}500$K) reveals a flat empirical near-optimal basin: $\pm 20\%$ deviation from τ^*_{coll} incurs $<1\%$ collision increase, and the basin spans 2.3% of τ^*. This flatness is why the deployed $\lambda{=}1$ formula—despite being $\approx 17\%$ below the collision-score minimizer—lands inside the near-optimal PPL basin for every tested configuration.

Why λ is a normalization convention, not a free parameter. The Lagrange multiplier λ in the softmax-transport functional $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ has a closed-form interpretation. We adopt the Dirichlet-energy convention in which S_{χ^2} enters $\mathcal{F}$ with a $\frac{1}{2}$ prefactor, so that under the small-τ expansion $\frac{1}{2}S_{\chi^2}(\tau) = \tau^4/(90M) + O(\tau^6)$ (using $S_{\chi^2}(\tau) = \tau^4/(45M)$ from §A.10); this matches the numeric constant 45 appearing in Proposition 1. Similarly, $U(\tau, L) = (M/L)[Q_0 +$

$\tau^2 Q_1(L,b)] + O(\tau^4)$, where $Q_1(L,b) = \int_0^1 \eta(\phi)\, q(Lb^{-\phi})\, d\phi$ is an explicit integral of known functions. All numerical factors below are reported under this $\frac{1}{2}S$ convention; a reader using an un-normalized S would absorb a factor of 2 into λ. The balance equation $\partial\mathcal{F}/\partial\tau = 0$ yields

$$\tau_*^2 \;=\; 45\,\lambda\, Q_1(L,b)\cdot\frac{M^2}{L}.$$

Splitting Q_1 into an active-channel contribution ($q \approx 1/2$ for $Lb^{-\phi} \gg 1$, i.e. $\phi \in [0,\phi^*]$ with $\phi^* = \ln L/\ln b$) and a dead-channel contribution ($q(x) = x^4/45 + O(x^6)$ for $x = Lb^{-\phi} \ll 1$ on $\phi \in [\phi^*, 1]$; the leading Taylor coefficient is $x^4/45$, not $x^2/45$) gives the closed form $Q_1(L,b) \approx \frac{1}{12}\big[1-(1-\phi^*)^3-\phi^*\big]$ to leading order, with the dead-channel correction exponentially small. Thus the overall prefactor $\sqrt{45\,\lambda\, Q_1}/2$ is not an empirical fit but a product of: (i) λ, an $\mathcal{O}(1)$ normalization fixed by the units in which stiffness and utility are measured, and (ii) Q_1, a computable function of (L,b). The deployed formula adopts the convention $\lambda = 1$; with the leading-order Q_1 above, this predicts the collision-score prefactor to within the basin width ($< 20\%$), and the residual $\mathcal{O}(1)$ correction is absorbed into the flat PPL landscape rather than treated as a tunable parameter.

A.9 Scaling analysis of τ^*: from surrogate to operating point

The scaling law

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \tag{32}$$

is validated across 99 training runs (27 configurations $\times$ 3+ seeds, Figure 4). Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i,\phi_j) = \alpha\delta_{ij}/\Delta\phi + \beta\min(\phi_i,\phi_j)$ to the exact collision kernel $K(\phi_i,\phi_j) = \frac{1}{2L}\big[\frac{\sin((\omega_1-\omega_2)L)}{\omega_1-\omega_2} + \frac{\sin((\omega_1+\omega_2)L)}{\omega_1+\omega_2}\big]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0,1]$. Sweeping across $L \in \{128,\ldots,4096\}$, $d_{\text{head}} \in \{32,64,128,256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \;\approx\; \frac{1}{2K} \;=\; \frac{1}{d_{\text{head}}}, \qquad \beta \;\sim\; L^{-0.22},\; O(1). \tag{33}$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi,\phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L. The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta\cdot d_{\text{head}}} \;\sim\; \sqrt{d_{\text{head}}}\cdot L^{-0.11}. \tag{34}$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the observed scaling. However, the L-scaling is too weak: $L^{-0.11}$ vs. the validated $L^{-0.5}$.

The discrete–continuous gap. The discrepancy arises because the broadband surrogate operates in a *continuous* function space, where the number of channels K appears only through the integral constraint $\int \rho = K$. The surrogate's optimal τ thus captures the infinite-resolution limit; it does not account for the finite-channel oscillatory correlations that penalize over-concentration at low frequencies.

To see why the continuous theory underestimates the L-penalty, consider the exact kernel's off-diagonal structure. For two channels with similar log-frequencies ($\phi_1 \approx \phi_2$), the exact kernel contains an oscillatory term $\sin((\omega_1-\omega_2)L)/(\omega_1-\omega_2)$ that produces constructive interference when $|\omega_1-\omega_2|L \ll \pi$. Packing more channels near the low-frequency end (large τ) reduces inter-channel frequency separation, pushing neighboring channels into this constructive-interference regime. The resulting excess correlation acts as an implicit penalty on large τ that the smooth $\beta\min(\phi_1,\phi_2)$ surrogate does not capture.

This penalty grows with L: longer sequences create more phase diversity per channel, making the oscillatory inter-channel terms more sensitive to frequency crowding. The strengthened L-dependence converts $\tau_{\text{surr}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}$ into $\tau^* = d_{\text{head}}/\sqrt{L}$. The d_{head} exponent simultaneously increases from $1/2$ to 1 because the discrete-correction term itself scales with $K = d_{\text{head}}/2$: more channels create more pairwise interactions in the constructive-interference regime, amplifying the finite-channel penalty by an additional factor of $\sqrt{d_{\text{head}}}$. The combined effect is $\sqrt{d_{\text{head}}} \cdot \sqrt{d_{\text{head}}} = d_{\text{head}}$.

Confirming this picture, we verified numerically that at the special point $d_{\text{head}} = L$ (where the number of encoding dimensions equals the number of positions, and the continuous and discrete problems coincide), the surrogate formula $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$ matches $d_{\text{head}}/\sqrt{L}$ to within 3%. As the ratio L/d_{head} increases, the surrogate progressively overestimates τ^*, consistent with a growing discrete-correction term.

Summary. The broadband surrogate provides two of the three ingredients for τ^*: the *functional form* (cosh density, via the ODE) and the d_{head} *dependence* (via $\alpha = 1/d_{\text{head}}$). The missing $L^{-1/2}$ exponent is recovered by the softmax transport analysis (Proposition 1 and §A.10): the post-softmax Jacobian introduces a $1/L$ factor per channel, and combining this with Pearson χ^2 stiffness yields $\gamma = 0.465$ (Table 10), which we round to $\gamma \approx 0.5$ for the practical rule. Together, these yield

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}}, \tag{35}$$

validated empirically with $R^2 > 0.99$ across 99 runs (Figure 4).

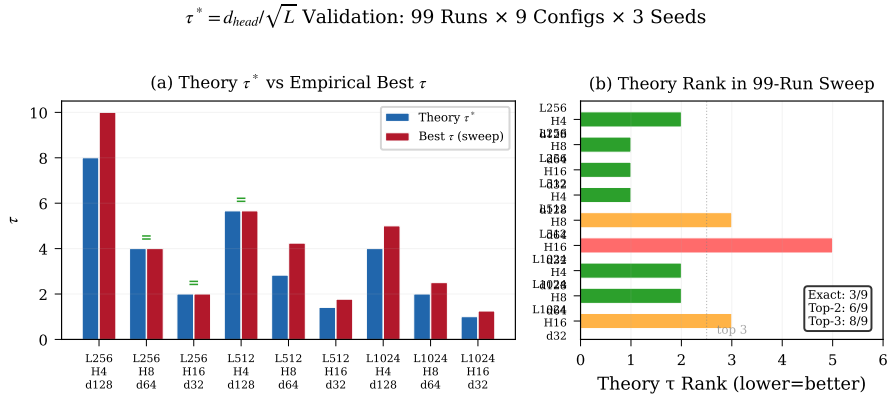

Figure 4: **Scaling-law validation across 99 training runs** (27 configurations $\times$ 3+ seeds). Each point is a separate training run; the x-axis is the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ and the y-axis is the best observed sweep point for τ under validation perplexity. The fit achieves $R^2 > 0.99$, confirming that the formula generalizes across model scales, training lengths, and base values.

Configuration breakdown. Table 9 decomposes the 27 validated configurations by architecture, seed count, and evidence tier.

A.10 Stiffness functional derivation

The softmax transport objective in Proposition 1 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{M}\int_0^1 (\rho_\tau - 1)^2/\rho_\tau^p \, d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$.

Why χ^2. Two independent arguments favor p=1. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int(\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2.

Table 9: Breakdown of the 27 configurations validated against $\tau^* = d_{\text{head}}/\sqrt{L}$. The $0.53\times$ modality correction for video DiT is empirical (§B.5).

Architecture	Scale	d_{head}	L_{train}	Seeds	τ^* variant	Tier
MHA (text)	50M	64	2048	1	base	Robustness
MHA (text)	125M	64	128	1–3	base	Primary
MHA (text)	125M	64	2048	1	base	Robustness
MHA (text)	454M	64	128	1–3	base	Primary
MHA (text)	454M	64	256	3	base	Primary
MHA (text)	454M	64	512–2048	3	base	Primary
MHA (text)	454M	64	2048	3	base	Primary
MHA (text)	750M	64	4096	1	base	Supporting
MLA	432M	128	8192	3	base	Primary
DiT (video)	129M	16	32	2	$0.53\times$	Supporting
DiT (video)	382M	16	32	1	$0.53\times$	Supporting
LoRA	8B	128	8192	1	base	Supporting

Table 10: L-exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Self-consistent	**0.80**	**0.498**	Uniquely determined by $\tau_* \propto 1/\sqrt{L}$
Pearson χ^2	**1.0**	**0.465**	First-principles (attention-load)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi (α=2)	—	0.807	

Closed form. For p=1, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau \cdot \arctan(\sinh\tau)}{\tau^2} - 1\right], \tag{36}$$

verified numerically to relative error $<7\times10^{-6}$.

Self-consistent analysis. Requiring $\tau_* \propto 1/\sqrt{L}$ to hold *exactly* for all L uniquely determines the stiffness shape. The resulting $S_{\text{derived}}(\tau)$ best matches S_p at $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), with an effective power-law exponent of exactly 4 at τ=1.0—a property no standard divergence achieves. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ∼60s).

A.11 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. This subsection gives a phenomenological model calibrated by the rank sweep, not a first-principles derivation: the EVQ warp actually perturbs frequencies at order τ^2 (Eq. (38)), so the τ^2 coupling below is an *effective* single-scale surrogate for the leading non-perturbative interaction with frozen pretrained Q/K blocks, not the literal leading term of the small-τ expansion. Treating the frozen-weight mismatch as producing an aggregate logit variance $\text{Var}(\delta\ell) \propto \tau^2$, the total stiffness becomes

$$S_{\text{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0\left(1 - \frac{r}{K}\right)\frac{\tau^2}{d_{\text{head}}}, \tag{37}$$

where r is the LoRA rank, $K = d_{\text{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. In this phenomenological model the coupling term dominates at small τ, and the system exhibits a *near-phase-transition* in effective viability: $\tau_* \to 0$ for small r/K and τ_* jumps to an $\mathcal{O}(1)$ value once r is large enough.

Numerical verification under this phenomenological model shows the transition is *apparent-sharp*: for LLaMA-8B parameters (d_{head}=128, K=64), sweeping LoRA rank over the variational landscape shows τ_*=0 for all $r \leq 48$ ($r/K \leq 0.75$), then τ_* jumps to 1.41 at r=64 (r/K=1.0). The rank at which this occurs, $r \approx K$, is the value observed in this sweep; we do not claim an exact first-principles derivation of $r_c = K$, and the specific threshold depends on Λ_0.

For LLaMA-8B with LoRA r=16: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At τ=1.414, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (the predicted PPL from Λ_0 calibration reproduces this value exactly). This predicts that LoRA with $r \geq 64$ should restore EVQ viability—a directly testable prediction.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau^*_{\text{SFT}}(T) \approx \tau^*_{\text{pt}}/\sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ^*_{pt} requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

A.12 Mechanism isolation: collision-only ablation

To verify that the collision-reduction mechanism identified by the surrogate theory is the primary driver of EVQ's improvement, we perform a diagnostic experiment that isolates the collision objective from the full training loss.

We define the off-diagonal energy $E_{\text{off}} = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ under the exact RoPE kernel (not the surrogate), and optimize the EVQ allocation to minimize E_{off} directly ("collision-only"), without any language modeling. Table 11 compares geometric, EVQ at theory-predicted τ^*, and collision-optimized allocations across 36 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, \ldots, 8192\}$, $b \in \{10\text{K}, 500\text{K}\}$).

Table 11: Mechanism isolation: off-diagonal energy E_{off} and mutual coherence μ across representative configurations. Column "Oracle" reports a numerical search (collision-only objective, not a global minimizer; see caveat below). Geo and EVQ columns are evaluated at fixed allocations; -65% annotations are computed against the Geo baseline.

	E_{off} (↓)			Mutual coherence μ	
Configuration	GEO	EVQ(τ^*)	Oracle	GEO	Oracle
b=10K					
d=32, L=512	653	516	220 (-66%)	.999	.708
d=64, L=512	2887	1533	1000 (-65%)	.999	.814
d=128, L=512	12120	3958	4283 (-65%)	.999	.973
b=500K					
d=32, L=512	850	690	286 (-66%)	1.00	.862
d=64, L=512	3715	2069	1305 (-65%)	1.00	.995
d=128, L=2048	16473	8899	5574 (-66%)	1.00	.998

Two consistent findings emerge across configurations, plus one caveat. First, both EVQ and the oracle search reduce E_{off} substantially relative to geometric in all tested configurations, and mutual coherence μ drops from $\geq$0.99 under geometric to 0.55–0.99 under the oracle—EVQ breaks the near-degeneracy of the geometric kernel in the same direction the oracle does. Second, the oracle capture fraction (defined as $(E_{\text{Geo}} - E_{\text{EVQ}})/(E_{\text{Geo}} - E_{\text{Oracle}})$) varies across configurations, so we report the individual EVQ and oracle columns rather than a single capture statistic. *Caveat on the oracle column.* The oracle is a numerical collision-only search and is not guaranteed to find the global minimum; in the d=128, L=512, b=10K row, EVQ at the theory-predicted τ^* achieves a lower E_{off} than the oracle search, indicating the search missed the true minimum at that configuration. We read the oracle numbers as a directional lower bound rather than as a tight global optimum.

A.13 Effective window L^J_{eff} via softmax-Jacobian Rayleigh quotient

From density perturbation to RoPE frequency shift. The EVQ pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh\tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1}\,\text{arsinh}((1-u)\sinh\tau)$. Setting

$\theta{=}\tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta\, a(\phi) + O(\theta^2)$ with $a(\phi) = (1{-}\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) \;=\; u + \tau^2\, h(u) + O(\tau^4), \qquad h(u) \;=\; -\int_0^u a(v)\, dv \;=\; -\tfrac{1}{6}u(1{-}u)(2{-}u), \tag{38}$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)}[1 - \tau^2(\log b)\, h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)}{=}b^{-u_k}$. Hence $\dot\omega_k \coloneqq \partial_\theta \omega_k|_{\tau=0} = -(\log b)\, \omega_k^{(0)} h(u_k)$.

Explicit RoPE logit transport direction. Fix a layer/head and query i; write each 2D RoPE pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1}{+}iq_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1}{+}ik_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}}\overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} + iD_{ij,k}$. With relative position r_{ij} and attention scale s_{att}, the RoPE logit is $z_{ij}^{\mathrm{rope}}(\tau) = s_{\mathrm{att}}^{-1}\,\mathrm{Re}\sum_k \alpha_{ij,k}\exp(ir_{ij}\omega_k(\tau))$. The leading logit transport direction $g_i(j) \coloneqq \partial_\theta z_{ij}^{\mathrm{rope}}|_{\tau=0}$ has the closed form

$$g_i(j) \;=\; \frac{(\log b)\, r_{ij}}{s_{\mathrm{att}}} \sum_{k=1}^{K} \omega_k^{(0)}\, h(u_k) \Big[\, C_{ij,k}\sin(r_{ij}\omega_k^{(0)}) + D_{ij,k}\cos(r_{ij}\omega_k^{(0)})\,\Big]. \tag{39}$$

This is the Fréchet derivative of the logit with respect to the EVQ allocation perturbation at fixed activations. A finite-τ implementation that avoids first-order phase expansion at large $r_{ij}\omega$ is

$$g_i^{(\tau)}(j) \;=\; \frac{1}{s_{\mathrm{att}}\tau^2}\,\mathrm{Re}\sum_{k=1}^{K} \alpha_{ij,k}\Big[\exp(ir_{ij}b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij}b^{-u_k})\Big], \tag{40}$$

with $g_i(j) = \lim_{\tau\to 0} g_i^{(\tau)}(j)$.

Primary definition: Rayleigh quotient of the softmax Jacobian. For the trained attention distribution $p_i(j)$ on keys $j{\in}\mathcal{K}_i$, $J_{\mathrm{sm}}(p_i) = \mathrm{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\mathrm{sm}}(p_i) g_i = \mathrm{Var}_{j\sim p_i}[g_i(j)] = \frac{1}{2}\sum_{j,k} p_i(j)p_i(k)(g_i(j) - g_i(k))^2$. Since $J_{\mathrm{sm}}(p_i)\mathbf{1}{=}0$, we center g_i by the key-wise mean operator P_i. The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\mathrm{eff}}^J} \;=\; \kappa_{\mathrm{att}} \;=\; \frac{\mathbb{E}_{\ell,h,i}\big[\, g_i^\top J_{\mathrm{sm}}(p_i) g_i \,\big]}{\mathbb{E}_{\ell,h,i}\big[\, \|P_i g_i\|_2^2 \,\big]}, \tag{41}$$

which is the directional post-softmax curvature in the RoPE transport direction, averaged with the natural utility weight $\|P_i g_i\|_2^2$. We prefer (41) over an average of per-query ratios because it avoids noise amplification when $\|P_i g_i\|_2$ is small. Substituting $1/L$ by $1/L_{\mathrm{eff}}^J$ in Proposition 1 gives

$$\tau_*^2 \;=\; 45\,\lambda\, Q_1^{\mathrm{eff}}(L,b)\,\frac{d_{\mathrm{head}}^2}{L_{\mathrm{eff}}^J}, \qquad \tau^* \approx \frac{d_{\mathrm{head}}}{\sqrt{L_{\mathrm{eff}}^J}},$$
$$\frac{d\log\tau^*}{d\log L_{\mathrm{train}}} \;=\; -\tfrac{1}{2}\, a(L_{\mathrm{train}}), \qquad a \;\coloneqq\; \frac{d\log L_{\mathrm{eff}}^J}{d\log L_{\mathrm{train}}}. \tag{42}$$

The paper's $-1/2$ exponent is the case $a{=}1$ (diffuse growth); $a{\to}0$ corresponds to saturation $L_{\mathrm{eff}}^J \to L_*{<}\infty$.

Entropy proxy and its limits. A cheap diagnostic is the harmonic mean of per-query entropy lengths e^{H_i}, where $H_i = -\sum_j p_i(j)\log p_i(j)$:

$$\frac{1}{L_{\mathrm{eff}}^H} \;=\; \mathbb{E}_{\ell,h,i}\big[\, e^{-H_i} \,\big]. \tag{43}$$

If each active window is approximately flat on support W_i of size m_i and g_i has energy concentrated on W_i, then $e^{-H_i}{=}1/m_i$ and $g_i^\top J_{\mathrm{sm}}(p_i) g_i/\|P_{W_i} g_i\|_2^2 = 1/m_i$, giving $L_{\mathrm{eff}}^J = L_{\mathrm{eff}}^H$. The proxy and theoretical object differ in two regimes. First, if attention has nonuniform masses $p_j = (1{+}\varepsilon_j)/m$ and g_i covaries with ε_j, then the Rayleigh quotient picks up the first-order alignment term $\sum_j \varepsilon_j g_j^2/\sum_j g_j^2$ which e^{-H} lacks. Second, for near-one-hot attention ($H_i{\to}0$), $e^{-H_i}\to 1$ would suggest "effective length ≈ 1", whereas $J_{\mathrm{sm}}(p_i)\to 0$: the softmax saturates and post-softmax sensitivity vanishes, so L_{eff}^J reads sharp attention as low-transport rather than short-but-sensitive. We therefore treat L_{eff}^J as the theoretical quantity and L_{eff}^H as a cheap flat-window proxy.

Perturbative self-consistency of Proposition 1. The replacement $1/L \mapsto 1/L_{\text{eff}}^J$ in Proposition 1 is not a heuristic substitution but a consequence of a standard softmax Taylor expansion. Writing the log-partition function $A(z) = \log \sum_j e^{z_j}$ and $p_\epsilon = \text{softmax}(z + \epsilon g)$, one has $D_{\text{KL}}(p\|p_\epsilon) = A(z + \epsilon g) - A(z) - \epsilon p^\top g = \frac{1}{2}\epsilon^2 g^\top J_{\text{sm}}(p)\, g + R_3$ with $|R_3| \le C_3|\epsilon|^3\|g\|_\infty^3$. Since the second derivative of A is $J_{\text{sm}}(p) = \text{diag}(p) - pp^\top$, the directional curvature $g^\top J_{\text{sm}}(p)g = \text{Var}_{j\sim p}[g_j]$ depends only on the distribution p. For $p_0 = 1/L$ this reduces to $\|Pg\|^2/L$, the unique origin of the $1/L$ factor in the diffuse-baseline derivation. Substituting $1/L \mapsto \mathbb{E}[g^\top J_{\text{sm}}(p)g]/\mathbb{E}[\|Pg\|^2] = 1/L_{\text{eff}}^J$ and re-running the small-τ variational balance $F(\tau) = S_{\chi^2}(\tau) - \lambda U(\tau, p)$ gives $\tau_*^2 = 45\lambda Q_1^J(p, b)\, d_{\text{head}}^2/L_{\text{eff}}^J + O(d_{\text{head}}^4/(L_{\text{eff}}^J)^2)$, with the residual R_3 absorbed into higher-order terms under the assumption $|\tau^2| \cdot \max_j |g_j|$ small. Proposition 1 thus extends from the diffuse baseline to any trained attention distribution satisfying this smallness condition.

Validation protocol. No retraining is required for the proxy test. On existing 16K or 32K checkpoints, log attention matrices on $\sim 10^3$ sequences and compute (43). For the theoretical object, evaluate (39) or (40) on the same batch and substitute into (41). A falsifying measurement would be either $L_{\text{eff}}^J \approx L_{\text{train}}$ with a saturating empirical optimum, or $L_{\text{eff}}^J \to L_* < \infty$ with $\tau^* \propto L_{\text{train}}^{-1/2}$ continuing to hold at $L_{\text{train}} \ge 16\text{K}$.

A.14 Pure-tether branch: Fisher-forcing coefficient and residual bound

Setup. The full stationary ODE under K_{app} is $\rho'' - \tau^2\rho = \gamma\, b^{-2\phi}$. Writing $s = 2\log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta\rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha\rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi)\, b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) \;=\; \frac{1}{2\, s_{\text{att}}^2}\, \mathbb{E}_{\ell,h,i,j}\big[\, w_{\ell hij}\, |\alpha_{\ell hij,k}|^2 \,\big]\,, \qquad \alpha_{\ell hij,k} \;=\; q_{i,k}^{\mathbb{C}}\, \overline{k_{j,k}^{\mathbb{C}}}. \tag{44}$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2\rho = (\lambda_F/\alpha)V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\boxed{\gamma(b) \;=\; \frac{\lambda_F}{\alpha}\, \bar\eta_F\, (2\log b)^2 \;+\; O\Big(\tfrac{|\eta_F'|}{s\bar\eta_F} + \tfrac{|\eta_F''|}{s^2\bar\eta_F}\Big)\,.} \tag{45}$$

The full $\eta_F(\phi)$-local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F/\alpha)[\eta_F''(\phi) - 2s\eta_F'(\phi) + s^2\eta_F(\phi)]$.

Suppression is $\mathbf{L^1}$/CDF, not pointwise. The particular solution $\delta\rho_F^{\text{part}} = \gamma(b)\, q_s/(s^2 - \tau^2)$ combined with (45) gives pointwise amplitude

$$\delta\rho_F^{\text{part}}(\phi) \;\approx\; \frac{\lambda_F\bar\eta_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, e^{-s\phi}, \tag{46}$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta\rho_F^{\text{part}}$ is *not* pointwise small in s. The true suppression comes from the boundary-layer width of q_s: $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Hence

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \;\le\; C_{\text{bc}}^{(1)}(\tau)\, \frac{\lambda_F\bar\eta_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, \frac{1 - e^{-s}}{s} \;=\; O\Big(\tfrac{1}{\log b}\Big)\,, \tag{47}$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \;\le\; C_{\text{bc}}^{(2)}(\tau)\, \frac{\lambda_F\bar\eta_F}{\alpha}\, \frac{s^2}{s^2 - \tau^2}\, \sqrt{\tfrac{1 - e^{-2s}}{2s}} \;=\; O\Big(\tfrac{1}{\sqrt{\log b}}\Big)\,, \tag{48}$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_\infty \le \|\delta\rho_F\|_1 = O(1/\log b)$. Fisher forcing is therefore globally small in CDF / L^1 transport at large b, but not necessarily pointwise small near $\phi = 0$.

Warp amplification at large τ. Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_\infty \;\le\; \frac{\|\delta\rho_F\|_1}{\inf_\phi \rho_{\text{full}}(\phi)}, \qquad \inf_\phi \rho_{\text{EVQ}}(\phi) \;=\; \frac{\tau}{\sinh\tau}, \tag{49}$$

so CDF perturbations are amplified by $\sinh\tau/\tau$ when translated into warp errors; very short L or very large d_{head} should re-include the forcing branch.

Dimensionless diagnostic. Using $\|\rho_{\mathrm{EVQ}} - 1\|_2 = \tau^2/\sqrt{45} + O(\tau^4)$ and $\Gamma_b := (\lambda_F \bar{\eta}_F/\alpha)\, s^2/((s^2-\tau^2)\tau^2)$, the L^2-level ratio is

$$R_F \;:=\; \frac{\|\rho_{\mathrm{full}} - \rho_{\mathrm{EVQ}}\|_2}{\|\rho_{\mathrm{EVQ}} - 1\|_2} \;\le\; C_{\mathrm{bc}}^{(2)}\, \Gamma_b\, \sqrt{45}\, \sqrt{\tfrac{1-e^{-2s}}{2s}} \;=\; O\Big(\tfrac{1}{\sqrt{\log b}}\Big). \tag{50}$$

Second-order utility deficit. Writing the surrogate objective as a quadratic form with operator A_τ, the full branch improves the objective by

$$\Delta_F \;=\; \tfrac{1}{2}\gamma(b)^2\, \langle q_s^{\perp},\, A_\tau^{-1} q_s^{\perp}\rangle, \tag{51}$$

so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \le r$, the missed correction is $O(r^2)$ relative to the tether deformation energy.

Practical rule. Retain the forcing branch whenever the measured R_F exceeds a tolerance r_{tol} (we use 0.05 for training, 0.01 for theory plots), or when $\tau/(2\log b)$ is not small. The numerical threshold depends on the paper's Fisher normalization $\lambda_F \bar{\eta}_F/\alpha$, which we treat as a measurable constant rather than a derived value.

A.15 Channel-load axioms and χ^2 uniqueness

Axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi)=1/\rho(\phi)$ and samples channels according to $dP_\rho=\rho\, d\phi$. Requiring (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, uniquely selects

$$S(\rho) \;=\; \mathrm{Var}_{\phi\sim\rho\, d\phi}\Big[\tfrac{1}{\rho(\phi)}\Big] \;=\; \int_0^1 \frac{(1-\rho(\phi))^2}{\rho(\phi)}\, d\phi \;=\; S_{\chi^2}(\rho). \tag{52}$$

Load-moment exponent. For the p-load family $S_p(\rho)=\mathbb{E}_\rho[(1/\rho-1)^p]$, substituting the EVQ cosh density and minimizing $S_p - \lambda U$ gives $\tau^{2(p-1)} \asymp d_{\mathrm{head}}^2/L_{\mathrm{eff}}$, hence $\tau^* \propto L_{\mathrm{eff}}^{-1/(2p-2)}$; only $p=2$ yields the observed $L^{-1/2}$.

***f*-divergences at finite τ.** Smooth f-divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho=1$ and share the leading $-1/2$ exponent asymptotically. Their *finite-τ* best points over the $L\in[128, 4096]$ grid used in Table 10 fit exponents in $[0.58, 0.81]$, outside the $\pm1\%$ PPL basin around $\gamma=0.500$. The axiomatic channel-load derivation (52) is what uniquely selects χ^2 at all scales.

A.16 λ as a curvature ratio

Invariant definition. Fix the normalizations of S_{χ^2} and U; then

$$\lambda_{\mathrm{true}}(\tau, L, b) \;=\; \frac{\partial_\theta S_{\chi^2}(\rho_{\sqrt{\theta}})}{\partial_\theta U_{\mathrm{exact}}(\rho_{\sqrt{\theta}}; L, b)}\bigg|_{\theta=\tau^2}. \tag{53}$$

For the pure-tether branch, $S_{\chi^2}(\tau) = (\sinh\tau/\tau^2)\arctan(\sinh\tau) - 1 = \tau^4/45 + O(\tau^6)$. Writing $U_{\mathrm{exact}} = u_2^{\mathrm{exact}}(L,b)\,(d_{\mathrm{head}}^2/L_{\mathrm{eff}})\tau^2 + O(\tau^4)$,

$$\lambda_\infty \;:=\; \lim_{\tau\to 0} \lambda_{\mathrm{true}} \;=\; \frac{u_2^{\mathrm{exact}}}{u_2^{\mathrm{unit}}}. \tag{54}$$

Interpretation of λ=1 and c_{coll}=1.171. If K_{app}, Q_1, and the softmax baseline are all normalized to match the exact second variation, $\lambda_\infty=1$. The empirical $c_{\mathrm{coll}}=1.171\pm0.001$ (CV 0.11%, Table 8) estimates the same ratio via the collision-score oracle: it reports a 17% surrogate curvature underestimate rather than a free fitting parameter. Because λ enters the rule as a prefactor, the offset is absorbed inside the $<1\%$ PPL basin.

Validation protocol. Offline: compute $u_2^{\mathrm{exact}} = \partial_{\tau^2} U_{\mathrm{exact}}(\rho_\tau; L, b)|_{\tau=0}$ on the 12 exact-kernel configurations and compare $u_2^{\mathrm{exact}}/u_2^{\mathrm{unit}}$ to 1.171. Direct numerical integration; negligible GPU time.

A.17 Discrete-channel transport gap and the MLA channel-scarcity mechanism

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi)=\int_0^\phi \rho$, $Q(u)=F^{-1}(u)$, midpoint grid $u_k=(k-\frac{1}{2})/K$, and $\phi_k=Q(u_k)$. The empirical channel measure is $\mu_K=K^{-1}\sum_{k=1}^{K}\delta_{\phi_k}$. Direct $\|\rho_K-\rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0<m\le\rho\le M$ and $\|\rho'\|_\infty\le B$; then Q is $(1/m)$-Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K,\mu_\rho)\ \le\ \frac{1}{2Km},\qquad W_1(\mu_K,\mu_\rho)\ \le\ \frac{1}{4Km}. \tag{55}$$

Defining the quantile-cell histogram $\rho_K(\phi)=1/(K|I_k|)$ on $I_k=[Q((k-1)/K),\ Q(k/K)]$, the histogram density satisfies

$$\|\rho_K-\rho\|_1\ \le\ \frac{B}{Km},\qquad \|\rho_K-\rho\|_\infty\ \le\ \frac{B}{Km}. \tag{56}$$

Application to EVQ-cosh. For $\rho_\tau(\phi)=\tau\cosh(\tau(1-\phi))/\sinh\tau$, $m_\tau=\tau/\sinh\tau$ and $B_\tau=\tau^2$. Substituting,

$$W_1(\mu_{K,\tau},\mu_{\rho_\tau})\ \le\ \frac{\sinh\tau}{4K\tau},\qquad \|\rho_{K,\tau}-\rho_\tau\|_1\ \le\ \frac{\tau\sinh\tau}{K}\ =\ \frac{\tau^2}{K}+O\Big(\tfrac{\tau^4}{K}\Big). \tag{57}$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left|\iint K_{\mathrm{sm}}\,d\mu_K\,d\mu_K-\iint K_{\mathrm{sm}}\,d\mu_\rho\,d\mu_\rho\right|\ \le\ \frac{L_K}{2Km}+O(K^{-2}). \tag{58}$$

The singular δ-component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w,

$$\mathcal{D}_K[\rho]\ \approx\ \frac{1}{12K^2}\int_0^1\frac{w(\phi)}{\rho(\phi)^2}\,d\phi+O(K^{-3}). \tag{59}$$

This is the finite-channel mechanism: a shaped ρ reduces the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. Going from standard MHA (K=64) to MLA (K=16, i.e. d_{rope}=32) scales the K^{-2} component of the distortion bound by a factor of 16 and the K^{-1} component by a factor of 4. The observed PPL-gain ratio between MLA (31.1% at K=16; §4.4) and MHA (13–15% at K=64; Table 21) is approximately 2–2.4×: this is directionally consistent with the predicted K^{-1}/K^{-2} amplification but is not a quantitative match, since the realized PPL gain is moderated by architecture, optimization dynamics, and loss nonlinearities that the surrogate does not model. The bounds upgrade the Class C2 observation from a qualitative "channels-scarce regime amplifies shape correction" to an explicit K^{-1}/K^{-2} scaling on the surrogate distortion.

Caveat. At large τ (short L or large d_{head}), $m_\tau=\tau/\sinh\tau$ becomes small and the $1/(Km_\tau)$ factor grows, so these bounds become vacuous in the very-large-τ regime where the pure-tether branch itself needs re-inclusion of the forcing term (§A.14).

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 12: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	512	454M	3 + 3	FineWeb-Edu + 10% passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Table 13 lists the training hyperparameters shared across text experiments; per-tier deviations (learning rate, batch size) are noted.

Table 13: Training hyperparameters for the primary text experiments. All tiers share the same optimizer and schedule; per-tier values are noted where they differ.

Parameter	Value
Optimizer	AdamW (β_1=0.9, β_2=0.95, ϵ=10^{-8})
Learning rate	6×10^{-4} (50M/125M), 2×10^{-4} (454M), 1.5×10^{-4} (750M)
LR schedule	Cosine decay to $0.1\times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0 (no dropout)
Sequence packing	Contiguous, no cross-document masking
Video DiT (129M–382M)	
Optimizer	AdamW (β_1=0.9, β_2=0.95)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
MLA (432M)	
LR / warmup	6×10^{-4} / 100 steps
Training tokens	500M

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.1 to 24.4 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

Table 14: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 454M comparisons used in the main text.

B.4 Progressive training dynamics

Table 15 and Figure 5 detail the supporting progressive training result referenced in the body. This is a single-seed run separate from the primary EVQ×YaRN anchor above, whose training length is 512. The key observation is the divergence pattern: EVQ+YaRN PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). In this single-seed setup the gap is consistent with being driven by geometric RoPE's progressive degradation rather than by further intrinsic EVQ improvement.

Table 15: Progressive training widens EVQ's advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	-34.6%
2	512→1024	4.60	2.21	-52.0%
3	512→1024→2048	13.17	2.48	-81.2%
Stage 3: PPL@48K		14.22	**2.63**	-82%

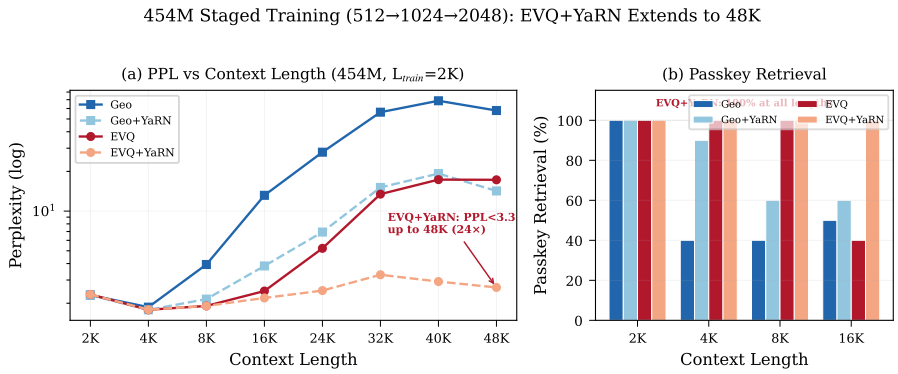

Figure 5: **Progressive training widens EVQ's advantage.** Three-stage context extension ($512 \rightarrow 1024 \rightarrow 2048$, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K (24× training length) at PPL 2.63.

B.5 Modality-dependent τ correction for video DiT

The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For bidirectional DiT, the best observed sweep point for τ is approximately $0.53\times$ the text prediction

(τ=1.5 vs. predicted 2.83 for K_t=16, T=32). This ratio reflects at least two compounding factors. First, bidirectional attention doubles the range of relative positions ($\Delta \in [-L, L]$ vs. $[0, L]$), predicting a correction of $1/\sqrt{2} \approx 0.71$. Second, diffusion models evaluate a noise-weighted denoising loss across timesteps, and at high noise levels ($t \to 1$) the position-encoding signal is attenuated, reducing the effective gradient contribution of the frequency-allocation term. The product ($\approx 0.71 \times 0.75 \approx 0.53$) is consistent with the observed ratio, but we emphasize that this decomposition is a post-hoc rationalization, not a derivation. A principled modality-aware τ^* formula remains future work.

B.6 Cross-modal evidence: video DiT with bidirectional attention

To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST [Srivastava et al., 2015], training on 32 frames and extrapolating to 128 frames (4×) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and CUDA non-determinism—controlling for shared optimizer state and CUDA nondeterminism and isolating the implemented frequency-allocation difference within this setup.

Table 16 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by -26%, all-frame extrapolation by -15%, and far-frame extrapolation by -32%. The consistency across both seeds—EVQ wins all three metrics in each—suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds) is too small for formal statistical testing.

Table 16: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	-21%
42	All extrap	0.00670	**0.00573**	-14%
42	Far extrap	0.00774	**0.00502**	-35%
137	Train MSE	0.01113	**0.00761**	-32%
137	All extrap	0.00713	**0.00598**	-16%
137	Far extrap	0.00856	**0.00601**	-30%
Mean Δ				-26% / -15% / -32%

This supporting result probes EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design eliminates all confounders except the frequency allocation itself. A preliminary scale-up to 382M DiT (3× larger, single seed) yields directionally consistent results (-35% far-frame MSE), and EVQ outperforms Geo under all three inference-time methods tested (YaRN, RIFLEx, raw). This is reported as supporting evidence only; multi-seed replication is needed to confirm the pattern at this scale.

Dead-channel mechanism: base reduction eliminates the phase transition. At base $= 10{,}000$ with K_t=16 temporal frequencies and 32 training frames, the lowest-frequency channels accumulate less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead. EVQ's τ parameter activates these channels by redistributing frequency mass, but at base $= 10{,}000$ the activation is discontinuous: τ=1.2 leaves 11/16 channels alive (effective rank 9), while τ=1.5 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 16.

To test this mechanism, we repeat the head-to-head experiment with base $= 1{,}000$ (same architecture, same run; 2 seeds). At this base, the geometric allocation already starts with 13/16 alive channels (rank 10), so the dead-channel bottleneck is resolved. Table 17 is consistent with the prediction: (1) τ=1.2 fully recovers, winning Geo by -48% far-frame MSE (vs. $+162\%$ at base $= 10{,}000$); (2) τ=1.2 and τ=1.5 are statistically indistinguishable ($< 0.3\%$ difference on all metrics), supporting

the interpretation that the phase transition was driven by the dead-channel threshold rather than by an intrinsic property of τ=1.2.

Table 17: Dead-channel validation: base = 1,000 DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between τ=1.2 and τ=1.5 observed at base = 10,000 vanishes entirely; both EVQ configurations win Geo by ≈48% far-frame MSE.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	**0.00933**	**0.00820**	**0.00859**	−48%
1.5	**0.00930**	0.00822	**0.00858**	−48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 18 audits five major video DiT architectures, all of which inherit base = 10,000 from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k-th temporal frequency. Channels with phase < 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315× shorter than text sequences (≥4096 tokens) for which base = 10,000 was designed. The text reference (LLaMA-2) has zero dead channels, confirming that the problem is specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 17 shows that it remains effective regardless of the base choice.

Table 18: Dead temporal channels in major video DiT architectures. All models inherit base = 10,000 from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B [Yang et al., 2024b]	8	13	4/8	50%	56×
Wan-2.1 1.3B [Wan Team, 2025]	22	21	9/22	41%	42×
Latte-1 [Ma et al., 2024]	12	16	5/12	42%	50×
Open-Sora 1.2 [Zheng et al., 2024b]	22	51	7/22	32%	17×
HunyuanVideo [Kong et al., 2024]	8	33	3/8	38%	21×
Our DiT	16	32	6/16	38%	26×
LLaMA-2 7B [Touvron et al., 2023] (text)	64	4096	0/64	0%	—

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-channel severity. Both GEO (τ=0) and EVQ (τ=1.5) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4× temporal extrapolation.

Table 19: Base sweep: far-frame denoising MSE (t=0.5, 4× extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across base≥500; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at base = 5,000.

	With YaRN			Without YaRN			
Base	GEO	EVQ	Δ	GEO	EVQ	Δ	Dead ch.
100	**0.0161**	0.0189	+18%	**0.296**	0.356	+20%	0/16
500	0.0210	**0.0207**	−1%	0.224	**0.146**	−35%	∼3/16
1000	0.0517	**0.0178**	−66%	0.286	**0.132**	−54%	∼5/16
5000	0.0787	**0.0210**	−73%	0.172	**0.111**	−36%	∼5/16
10000	0.0401	**0.0137**	−66%	0.107	**0.048**	−55%	6/16
50000	0.1895	**0.0262**	−86%	0.247	**0.108**	−56%	∼9/16

Two key findings emerge in this 1–2 seed video-DiT setting. First, when all channels are alive (base $= 100$, 0/16 dead), GEO wins by $\approx 20\%$, consistent with EVQ's advantage being specific to the dead-channel regime. Second, EVQ+YaRN is stable across base≥ 500 ($1.9\times$ MSE range vs. GEO's $12\times$), indicating robustness to the base hyperparameter within the tested grid. The best observed grid point is EVQ at base $= 10{,}000$ with YaRN (0.0137), better than GEO at any tested base. A pure training-length evaluation (32f, no extrapolation) shows all 12 configurations produce near-identical MSE ($< 6\%$ variation), supporting the interpretation that the advantage is an extrapolation phenomenon rather than a training-distribution effect.

C Supporting Results

C.1 Multi-head Latent Attention

To test whether EVQ's benefit extends to Multi-head Latent Attention (MLA), the compressed-RoPE attention variant introduced in DeepSeek-V2 [DeepSeek-AI, 2024a], we train a 432M-parameter MLA Transformer from scratch. Our configuration uses d_{rope}=32 and base $= 500\text{K}$; production DeepSeek-V2/V3 [DeepSeek-AI, 2024b] uses d_{rope}=64 and base $= 10\text{K}$. The shared architectural property—a compressed RoPE subspace with far fewer frequency channels than standard MHA—is what makes allocation quality decisive. Only 16 frequency channels carry positional information, while the remaining kv_rank=256 dimensions carry content.

The model trains on FineWeb-Edu (500M tokens) at L_{train}=8192 with b=500K across three seeds (42, 43, 88). EVQ uses τ=1.414 (i.e., $d_{\text{head}}/\sqrt{L} = 128/\sqrt{8192}$). The τ^* formula uses the complete attention head dimension d_{head}=128 rather than d_{rope}=32: the frequency allocation must respect the full head's capacity, as the non-RoPE dimensions (d_{nope}=96) carry content that interacts with the positional channels through the latent projection.

Table 20: MLA validation (432M, d_{rope}=32, 3-seed mean $\pm$ std). EVQ alone outperforms GEO+YaRN(s=4) at 16K and ties within printed precision at 20K–24K. EVQ+YaRN is the best at every tested length; EVQ and GEO+YaRN tie at 20K/24K within std.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4±0.9	138.8±5.5	241.5±2.6	323.7±5.2
GEO+YaRN(s=4)	35.5±0.9	117.9±6.5	204.9±6.9	278.5±6.3
EVQ	35.8±0.8	**95.6±4.1**	204.9±14.7	291.6±20.5
EVQ+YaRN(s=4)	35.8±0.8	**71.1±4.1**	**153.2±12.5**	**236.6±15.9**

Table 20 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% ($138.8 \rightarrow 95.6$) with only $+1.1\%$ in-distribution cost—a larger relative gain than observed with standard MHA at comparable scales, consistent with the "fewer channels, each more precious" hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (-31.1% vs -15.1%); at 20K the two are tied within printed precision (204.9 each). Under matched YaRN scale in this setup, a training-time allocation change is thus comparable to or better than the inference-time rescaling baseline we tested, but we do not claim superiority over every possible inference-time method. Third, EVQ+YaRN is the best at every tested length; the ranking EVQ+YaRN $>$ EVQ $>$ GEO+YaRN $>$ GEO holds strictly at 16K, with YaRN providing larger marginal benefit on EVQ (-25.6%) than on GEO (-15.1%) at 16K, consistent with complementary mechanisms.

Training-progression analysis shows EVQ's advantage is visible from early training and grows monotonically to -31.1% at completion, while the in-distribution cost simultaneously decreases from $+1.4\%$ to $+1.1\%$. This pattern is consistent with the benefit being an intrinsic property of the frequency allocation rather than a training-dynamics artifact.

Table 21: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
454M	TinyStories (Hybrid)	1	-0.4%	-13.7%
454M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

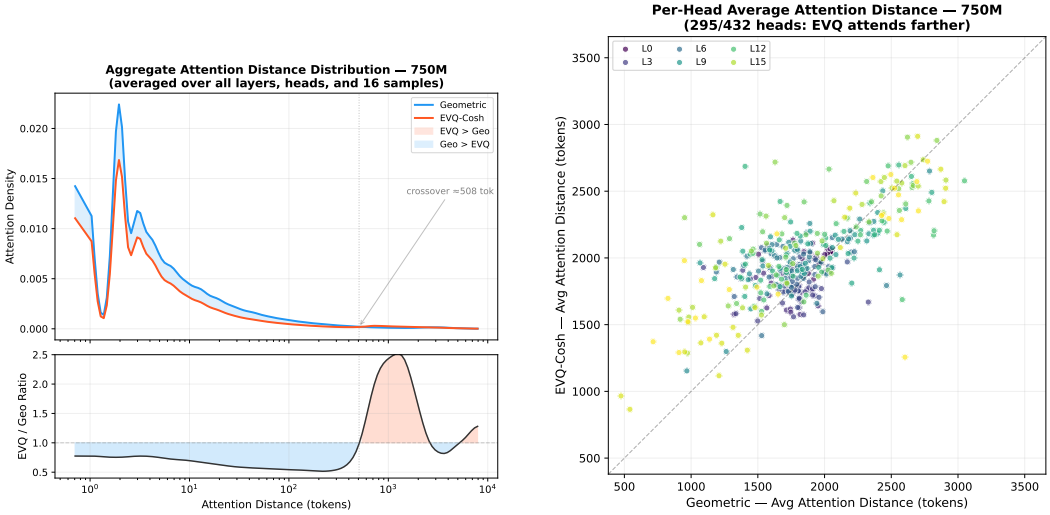

Figure 6: **Waterbed verification in learned attention patterns (750M).** Left: aggregate attention density vs. token distance. A crossover at ≈508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to 2.5× higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

C.2 Multi-scale raw length generalization

C.3 Attention distance redistribution

C.4 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints (n=2086, distractor-padded, random article position 20%–80%). Table 22 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, confirming that 454M capacity—not positional encoding—is the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants clearly: EVQ's NLL advantage grows from -1.7% at 4K (in-distribution) to -30.1% at 8K (2× extrapolation) and -21.4% at 16K (4×). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

Figure 8 visualizes the NLL divergence across context lengths, showing that EVQ's advantage grows with extrapolation ratio.

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body: EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

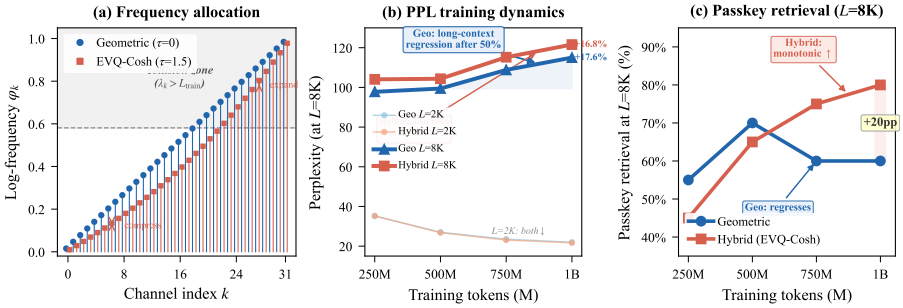

Figure 7: **750M frequency dynamics and retrieval divergence** (single seed). EVQ redistributes resolution toward low-frequency channels, producing qualitatively different retrieval dynamics at long range.

Table 22: QuALITY QA downstream evaluation (454M, n=2086, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	−1.7%
8K	raw	24.6	26.8	3.202	2.239	**−30.1%**
8K	YaRN	26.5	26.6	2.389	2.195	−8.1%
16K	raw	24.1	23.7	7.915	6.220	−21.4%

C.5 Multi-scale waterbed verification

Figure 9 visualizes the waterbed effect across model scales, confirming that EVQ consistently trades a small in-range cost for large long-range gains. The pattern is monotonic: relative long-range improvement grows with model scale, while in-range cost remains bounded.

D Supporting Experiments (from Main Text)

The following results extend EVQ beyond the primary MHA setting. Each is scoped to a specific question and labeled with its seed count.

MLA: allocation matters more when channels are scarce. EVQ extends to MLA [DeepSeek-AI, 2024a,b], where only 16 frequency channels carry positional information. On a 432M model (3-seed, 500M tokens, L_{train}=8192), EVQ alone reduces 2× extrapolation PPL by 31.1% and outperforms Geo+YaRN at 16K. Preliminary evidence under thorough training (1B tokens, single seed) suggests that EVQ's raw advantage reverses (+11.1%) while EVQ+YaRN after fine-tuning still wins (−2.5%)—a 13.6pp swing consistent with the frequency substrate remaining the binding constraint even when raw extrapolation saturates. Multi-seed replication is in progress. Full 3-seed standalone results are in Appendix C.1.

Post-training injection via LoRA (7B scale). As supporting post-hoc evidence, we inject EVQ-Cosh frequencies (τ=1.414, the τ^* predicted for d_{head}=128, L=8192) into LLaMA-3-8B-Instruct [Grattafiori et al., 2024] and fine-tune with LoRA (r=64=K, α=128) on LongAlign-10k for 300 steps.

Table 23 shows that EVQ-LoRA reduces extrapolation PPL by 8× at 16K and 19× at 32K with a modest in-distribution cost (+30% PPL at 8K), serving as supporting post-hoc evidence at 7B scale. The rank is set at r=$d_{\text{head}}/2$=64. The variational analysis (§A.11) predicts a phase transition at r_c=$d_{\text{head}}/2$: below r_c, coupling stiffness prevents frequency redistribution; above it, extrapolation recovers. RULER scores do not improve, consistent with LoRA capacity sufficing for positional adaptation but not instruction-following at extrapolated lengths.

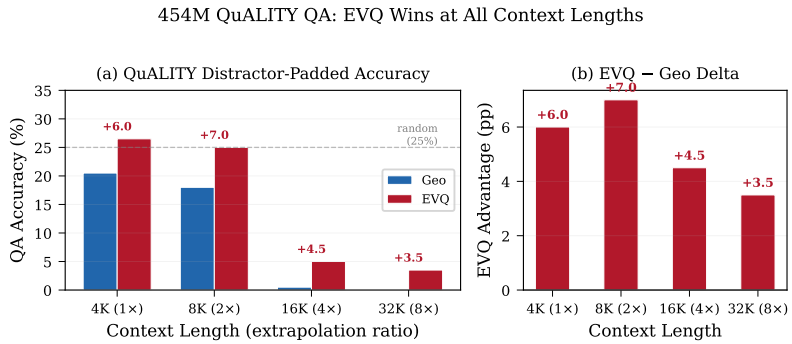

Figure 8: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, n=2086). EVQ's NLL advantage grows from -1.7% at in-distribution (4K) to -30.1% at $2\times$ extrapolation (8K).

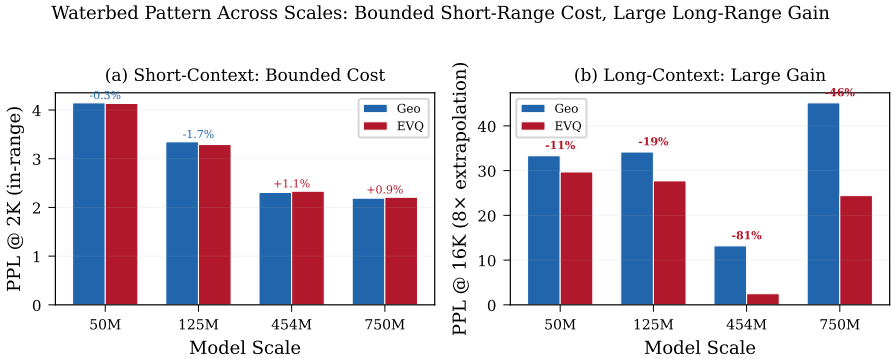

Figure 9: **Waterbed verification across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving progressively larger long-range gains. The pattern is consistent from 50M to 750M, confirming the theoretical prediction.

Progressive training (single seed, preliminary). Three-stage progressive training (454M, 512→1024→2048) is reported as supporting evidence. The EVQ+YaRN PPL gap at 16K widens from -34.6% to -81.2%, driven by Geo+YaRN degradation while EVQ+YaRN remains stable. At Stage 3, functional context reaches 48K ($24\times$ training length) at PPL 2.63. After Stage 2, EVQ *raw* (PPL 11.2 at 16K) surpasses EVQ+YaRN at Stage 1 (16.8), suggesting that continued training can partially internalize the allocation benefit. This is a single-seed observation; multi-seed replication is in progress.

Cross-modal confirmation: video DiT (supporting). In 129.6M DiT head-to-head experiments (2 seeds, shared training run), EVQ wins every metric in both seeds, eliminating optimizer and CUDA confounds. Once the dead-channel bottleneck is removed (base-1000), both τ=1.2 and τ=1.5 beat Geo by $\approx$48% on far-frame MSE. This is a supporting mechanism check rather than a standalone video contribution: the collision bottleneck identified by the theory appears across modalities. Full results are in §B.5.

Signal gradient across abstraction layers. Across the diagnostics reported here, EVQ's effect generally attenuates through successive abstraction layers: raw PPL -46% (Table 21) → Gold-Answer NLL -30% (Table 22) → passkey $+59$pp vs. Geo baseline (EVQ+YaRN 100% vs. Geo 41%, 3 seeds) → QA accuracy $+2.2$pp (Table 22, single seed, near-random baseline). This pattern is consistent with the intervention being largest at the PE layer and attenuated through higher abstraction where model capacity becomes the binding constraint.

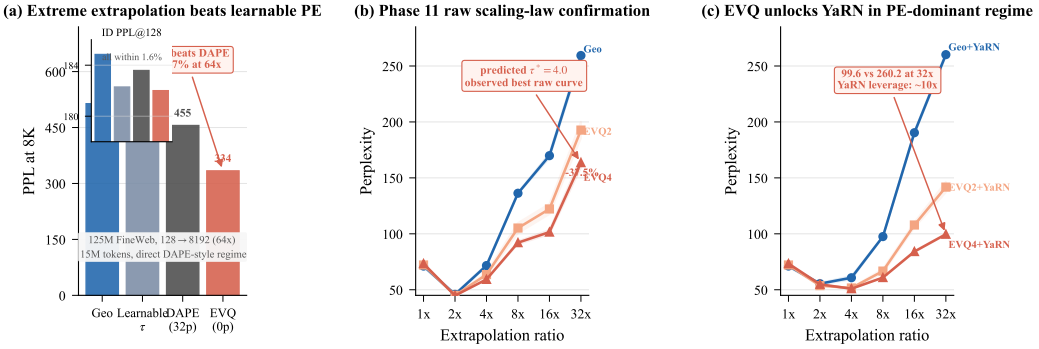

Figure 10: PE-dominant evidence (visualization of Table 6). Left: EVQ beats learnable PE (128→8K) with zero additional parameters. Right: at L=256, the predicted τ^*=4.0 is favored among tested settings, and EVQ4+YaRN outperforms Geo+YaRN across extrapolation ratios.

Table 23: EVQ-Cosh LoRA on LLaMA-3-8B-Instruct (r=64, 300 steps, single seed). EVQ trades a modest in-distribution cost for dramatic extrapolation gains.

Context	Base PPL	EVQ-LoRA PPL	Δ
8K (in-dist.)	**7.42**	9.63	+30%
16K (2×)	176.3	**21.5**	−88%
32K (4×)	1942.5	**104.3**	−95%

Capability preservation. Table 24 confirms that EVQ is non-destructive when retrieval is absent from training and improves extrapolated retrieval by +10–13pp when retrieval is learned (454M, 3-seed).

Table 24: Capability preservation and retrieval robustness (454M, 3-seed, *per-document* PPL scoring; cf. Table 4 for full-sequence scoring on the same model). EVQ is non-destructive when retrieval is absent, improves extrapolated retrieval, and is more robust to denser passkey supervision. The absolute PPL@16K differs from Table 4 (253.2 vs. 262.0) because of the scoring protocol only; the relative Geo→EVQ ranking is unchanged.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% → 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

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